



The impact of macroeconomic news sentiment on interest rates

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ABSTRACT

We provide evidence that sentiment extracted from articles related to interest rates, inflation, and the labor market has the ability to explain short-term interest rate movements that cannot be accounted for by professionals' and consumers' expectations. Additionally, sentiment can pin down two short rate regimes that are correlated with the business cycle. By combining these results with a yield curve model, we find that market sentiment has a statistically significant negative effect on the short end of the yield curve and a positive effect on the slope. We also show that sentiment improves the out-of-sample forecast accuracy of short-term yields.

1. Introduction

The perception that asset prices and business cycles are driven by expectations which do not necessarily stem from rational probabilistic calculation, is nothing new. In his publication, Keynes (1936) states that human beings' "innate urge to activity and rational selves", which is influenced by "motive on whim, sentiment or chance", makes the "(economic) wheels go round". Because of the challenge to model what Keynes (1936) called "animal spirits" (irrational human behavior), scholars ignored this issue for a large part of the last century.

Thanks to recent advancements in data analysis methods, scholars now have better tools to incorporate sentiment, i.e. the disposition of market investors towards financial markets, into their theoretical and empirical models. This is essential for several reasons: Firstly, incorporating sentiment can enhance economic understanding of various market phenomena that cannot be explained by rational expectations alone. Second, market sentiment plays a crucial role in driving trends, movements, and risks in asset prices (see, for example, Tetlock (2007) and Antweiler and Frank (2004)). Lastly, sentiment analysis can provide early warning signs of shifts in market conditions and can help investors stay ahead of the curve.

Despite the increasing amount of research on the relationship between financial markets and sentiment, most studies have focused on the equity market. This lack of attention to the fixed income market is somewhat surprising considering the overall value within the fixed income market, its high spillover effects on the real economy, and the large exposure that governments, corporations, financial institutions, and other organizations have to interest rates.

This paper aims to highlight the benefits of integrating sentiment analysis into the yield curve. To build our model, we start with a short rate analysis based on the Taylor rule that incorporates sentiment related to interest rates, inflation, and unemployment rate. We then extend this linear model to account for non-linearities in the form of sentiment-based regime-switches. Finally, we expand the analysis to the yield curve by assuming a term structure model of the Nelson–Siegel family and demonstrate the benefits of allowing for interactions between sentiment and yield curve factors.

The empirical results strongly favor the inclusion of sentiment in the short rate process together with inflation and unemployment. We find that sentiment related to interest rates news have a statistically significant effect on the 3-months Treasury yield. Furthermore, we show that this significance is maintained after controlling for professionals' and consumers' expectations, implying that the predictive ability of sentiment goes beyond them.

Several studies have confirmed that the short rate dynamics is subject to different regimes (see, for example, Hamilton, 1988). In order to exploit this stylized fact we estimate a Threshold Autoregressive Process (TAR) model. We are able to pin down two regimes which are characterized by very high and middle-to low sentiment scores. Fig. 1 shows the path of the U.S. 3-month Treasury yield for the two different regimes. To better relate the state of the economy and the sentiment regimes we also include the periods of economic contraction according to the National Bureau of Economic Research (NBER).

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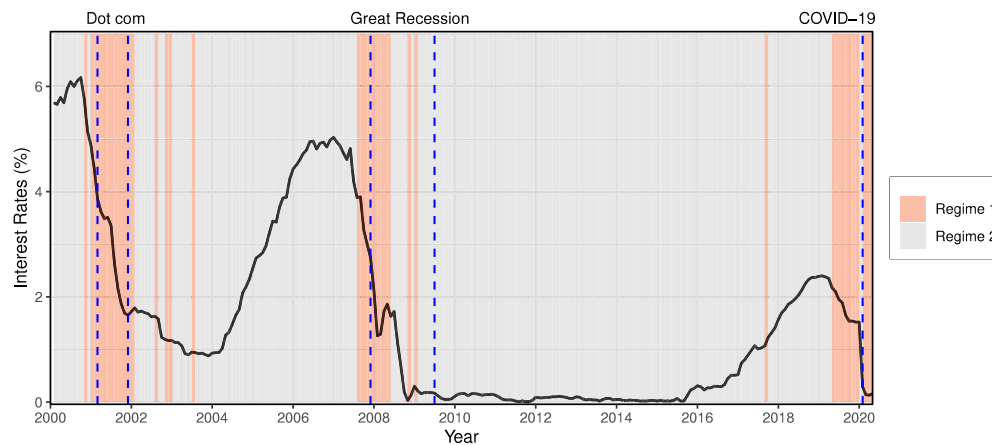


Fig. 1. U.S. 3-month treasury yield: Regime changes and economic contractions.

The figure shows monthly development of the U.S. 3-month Treasury yield from January 2000 to May 2020. The red shaded area represents the period characterized by higher sentiment scores, while the gray shaded area represents lower sentiment scores. The dashed blue lines indicate the periods of economic contractions according to NBER, which are indicated above the graph for reference.

Fig. 1 exhibits a total of three big switches to regime 1 (high sentiment score), all of which happened before the start of an economic contraction.¹ Notably, the first two switches preceded the start of the recession by 4 months, while the third switch occurred 9 months before. This suggests that regimes based on sentiment may hold predictive power over economic crises. However, while at the beginning of 2002 there was a switch back to regime 2 only after the end of the crisis, in the Great Recession there was a switch back to regime 2 13 months before the crisis was over. This may be due to a precipitated adjustment in sentiment caused by the brief reversal of the short rate at the beginning of 2008. Although the analysis is not an out-of-sample exercise, it indicates the possible use of this approach as an early warning signal for the start of a contraction period.²

To expand the analysis to the yield curve in slightly different theoretical frameworks, we work with two different models of the Nelson–Siegel Family. The first model, introduced by Diebold et al. (2006), accounts for interactions between sentiment and three yield curve factors. The second model is a four-factor no-arbitrage yield curve model, which allows sentiment to affect the yield curve through the short-term interest rate only. These choices enable us to evaluate the effectiveness of sentiment in explaining and forecasting the yield curve in different contexts. We explore both models using one and two sentiment regimes.

The statistical results strongly favor interactions between sentiment and yield curve factors. We find that interest rates sentiment has a significant negative impact on the short end of the yield curve, while positively affecting the slope. Impulse response functions further suggests that the effects of sentiment shocks on the short end of the yield curve can persist for several months. Overall, we interpret these results as follows: The perception of highly positive news anticipates a drop in the short-term interest rate. Moreover, the short-term interest rate will fall relative to long-term yields, causing the slope to increase. A steeper yield curve is typically associated with higher unemployment rates. As we will show, a regression of unemployment rate on interest rates sentiment is in line with this interpretation.

Finally, an out-of-sample forecast demonstrates that sentiment improves forecasting accuracy of yields. In total, the inclusion of the two-regime sentiment model decreases the 1-month forecasting error by around 2% relative to a Random Walk, 3% to the AR(1) model, and 10% to models using solely macroeconomic data and consumers'

¹ We define a big switch as a switch to regime 1 that lasts for longer than one quarter.

² Due to the small number of recession periods in our sample, we prefer to leave a rolling-window analysis for future research.

expectations. Although the improvement relative to the Random Walk might sound low, the Random Walk performs on average better than several models including the Nelson–Siegel, which is regarded as a good forecasting model (see, for example, Diebold and Rudebusch 2013).

This paper contributes to a vast literature that aims to understand financial markets through the lens of non-fundamental information. Our paper addresses one relatively unexplored area of the sentiment literature, namely the fixed income market. As detailed in Section 2, prior investigations into the effects of sentiment on the fixed income market have been limited in scope, primarily concentrating on credit spreads. Furthermore, the few existing studies that have examined the impact of sentiment on yields have primarily taken an empirical and linear approach and did not evaluate the out-of-sample forecasting performance of sentiment. In contrast, our approach integrates empirical sentiment analysis with a more theoretical foundation. We incorporate no-arbitrage conditions into one of our term structure models and establish a connection between our two sentiment-based regimes and the business cycle. This enhances the economic interpretation of our results and brings them closer to reality. Additionally, our paper stands out as the first to provide empirical evidence regarding the out-of-sample predictive ability of sentiment on U.S. Treasury yields, challenging traditional yield curve models that rely solely on state macroeconomic variables or expectations derived from surveys.

The remainder of this paper is organized as follows: Section 2 reviews the existing literature related to this work. Section 3 is devoted to the measurement of sentiment and its analysis in relation to various macroeconomic variables, professionals' and consumers' expectations, and the yield curve factors. Section 4 presents the results on the relationship between sentiment and the short rate. Section 5 introduces the yield curve models and their empirical findings. In Section 6 we address some limitations of our study. We conclude in Section 7.

2. Literature review

In recent years, scholars have focused their research on the development of methods for extracting sentiment data from raw text and applying them to econometric analysis. See, for example, Algaba et al. (2020) and Renault (2020) for a survey of advancements in this field. Based on these methods, a growing number of published studies have aimed to explain and forecast asset price movements in the equity market using sentiment; see, among many others, Antweiler and Frank (2004), Tetlock et al. (2008), Bollen et al. (2011) and Heston and Sinha (2017).

In contrast, studies of sentiment and interest rates are relatively limited. Various papers have examined the relationship between sentiment and central bank communication, with a prominent example

being [Lucca and Trebbi \(2009\)](#). A major focus of sentiment studies unrelated to central banking is bond spreads. For example, [Spyrou \(2013\)](#) examines the effects of investor sentiment on yield spreads during the Great Recession and finds statistically significant results. In addition, [Nayak \(2010\)](#) explores the impact of investor sentiment on corporate bond spreads and finds that sentiment has a substantial impact on corporate bond spreads and “mispricings and systematic reversal trends are very similar to those for stocks”.

Studies analyzing the interaction between macroeconomic news and interest rates have mainly focused on macroeconomic data releases. For example, [Mitra and Mitra \(2011\)](#) study the impact of information releases on market-level uncertainty in interest rates and foreign exchange futures markets. Similarly, [Edison \(1997\)](#) shows the explanatory power of inflation- and employment-related releases for Treasury yields, and [Goldberg and Leonard \(2003\)](#) find that the largest moves in yields are associated with U.S. announcements on labor market conditions, real GDP growth, and consumer sentiment. These studies highlight the importance of inflation and labor market news on movements in Treasury yields.

This paper is closely related to the study conducted by [Gotthelf and Uhl \(2019\)](#). They show that sentiment helps explaining and predicting yield curve factors and that sentiment affects the yields in a direction opposite to the slope factor. The primary difference between this study and our work, aside from the methodologies employed, is that we use sentiment scores derived from three different topics and examine them individually. In contrast, [Gotthelf and Uhl \(2019\)](#) include one single sentiment score derived from a mixture of topics including political, debt, and monetary policy news.

This paper also contributes to the existing literature on term structure models, with a particular focus on the Nelson–Siegel family. We chose to work with the Nelson–Siegel models because they are preferred over popular affine models as a forecasting tool (see [Diebold and Li 2006](#)). Furthermore, recent advancements in the Nelson–Siegel models by [Christensen et al. \(2011\)](#) and [Li et al. \(2011\)](#) have enhanced their economic interpretability.

Our study employs the Arbitrage-free Short Rate based Model (AFSR), which was first introduced by [Nyholm \(2018\)](#) to match the coefficients of [Söderlind and Svensson \(1997\)](#). However, unlike the latter model, the AFSR includes only one shape parameter, avoiding the issue of multicollinearity, as pointed out by [Pooter \(2007\)](#).

3. Data and summary statistics

3.1. Constructing news sentiment

In line with [Algaba et al. \(2020\)](#), we define news sentiment as the disposition of market investors towards interest rates, inflation, and unemployment rate via news from the financial media. The global macro dataset provided by RavenPack News Analytics is well-suited to measure this type of sentiment. RavenPack classifies news articles using multiple algorithms, which output data in the form of numerical news scores. There is strong empirical evidence that RavenPack News Analytics has explanatory and predicative power in market direction, volume, and volatility (see [RavenPack News Analytics \[RPNA\], 2020](#)). Furthermore, their sentiment scores have been shown to add value to forecasting German Bund yields, different European sovereign spreads and CDS spreads (see [Erlwein-Sayer, 2017, 2018](#); [Yang et al., 2020](#)).

The output chosen to reflect sentiment is the Event Sentiment Scores (ESS), which varies between 0 and 49 for negative sentiment and 51 and 100 for positive. 50 indicates neutrality of the news. The ESS is created by the RavenPack’s News Analytics Algorithm which applies NLP methods to news published by Dow Jones Newswires, Wall Street Journal, Barron’s, and Market Watch. This algorithm was built using scores derived from a collection of surveys in which finance and economics professionals rated entity-specific events as conveying positive or negative sentiment (see [Credit Suisse, 2021](#)). The strength of

ESS is further influenced by other factors such as the polarity of words used by the author to describe the event.

To construct the sentiment variable, we first extract the article-based ESS related to each macroeconomic variable using the keywords presented in [Table 1](#). We aggregate the ESS scores by taking the mean of all articles in a given month for each of the macroeconomic variables. Formally, our sentiment score measure is given by:

$$S_h(t) = \frac{1}{N} \sum_{j=1}^N \text{ESS}(k_{j,h}) \mathbb{1}_{\{Month(k_{j,h})=t\}}, \text{ for } h = i, \pi, u,$$

where $k_{j,h}$ represents one news item about h , and i , π , and u denote interest rates, inflation, and unemployment rate, respectively. N is the total number of observed articles in a month, and t refers to the observed month.

[Table 1](#) provides information on the keywords and descriptive statistics of the sentiment scores. The distribution of sentiment varies depending on the underlying variable. For example, the sentiment associated with inflation has a lower serial correlation than that of interest rates and unemployment. Furthermore, interest rates and inflation are positively-skewed, whereas unemployment is negatively-skewed. Notably, the mean inflation sentiment is less than 50, indicating that, on average, news articles discussing inflation are rated negatively.

To demonstrate the quality and accuracy of our sentiment scores, we provide an illustrative example. We compare the sentiment scores of Ravenpack with scores estimated using the bag-of-words method, which is a common approach in the economics literature (see, for example, [Gentzkow et al., 2019](#)). [Table 2](#) presents six articles, each scored by Ravenpack and three widely used lexicons in the economics literature: Loughran–McDonald (LM), psychological Harvard-IV dictionary (GI) and Hu and Liu (HL) (see, for example, [Gentzkow et al. \(2019\)](#)).

The accuracy of Ravenpack’s sentiment scores is evident in the provided examples, with all scores correctly classified according to human sentiment classification. In contrast, each lexicon fails to estimate the polarity of at least one article correctly. For instance, HL assigns neutrality to news regarding low unemployment, while LM and GI score news suggesting that inflation is not a significant concern as negative. In summary, we conclude that Ravenpack’s algorithm captures the true polarity of news articles better than standard approaches, making it a reliable source of sentiment scores.

3.2. Interpreting news sentiment

Interpreting sentiment scores poses a significant challenge for sentiment analysts, especially when relying on bag-of-words methods. As illustrated in [Table 2](#), the two lexicons classify news articles about inflation as negative, even though the headline indicates that inflation is not a concern. This discrepancy may arise because the word “inflation” has a negative score in the lexicon.

Fortunately, Ravenpack is able to capture the context of the article, as demonstrated in our previous example. Therefore, we can interpret the sentiment score as a measure of the subjective perception of investors towards the underlying variable. For instance, if investors view current news articles on inflation positively, the sentiment will be positive.

Furthermore, sentiment scores are generated based on investors’ emotions towards the variables discussed in the news articles, rather than the market consequences of such news. For instance, an article with positive news on unemployment rate may imply negative news to the market, as central banks may opt to raise interest rates. Nevertheless, because the article is about unemployment rate, the unemployment rate sentiment will be positive.

Interpreting sentiment related to interest rates is challenging because the same news on interest rates may have different implications for savers and borrowers. However, since we use an aggregate sentiment score, we can reasonably assume that interest rates sentiment reflects how the majority of investors perceive news. As we will discuss

Table 1
Summary statistics of sentiment scores.

	Interest rates	Inflation	Unemployment rate
Keywords	interest-rate, Treasury-bill (-note, -bond) sovereign-debt, government-budget	inflation, deflation consumer-price-index	unemployment, employment jobless-claims, non-farm payrolls
Number of News	26,759	9,911	16,831
Mean	50.03	42.06	55.99
Std. Deviation	10.67	7.20	10.88
Skewness	0.32	0.24	-0.60
Kurtosis	3.29	3.73	3.37
ACF(1)	0.68	0.18	0.48

The table shows a summary of sentiment scores based on interest rates, inflation, and unemployment rate. The sentiment scores were extracted using specific keywords from January 2000 to May 2020.

Table 2
A comparison of Ravenpack and other Lexicons.

Headline	Ravenpack	LM	GI	HL	True sentiment
Fed's Mester: Fed Will Need To Raise Rates Three Times In 2018, 2019	-0.78	0.000	-0.016	-0.005	-
Federal Reserve Will Make Interest Rate Cut - Economists	0.78	-0.022	0.000	0.004	+
Fed's Quarles Says Low U.S. Inflation Readings Are Not A 'great Concern'	0.74	-0.007	-0.021	0.007	+
MARKET TALK: US PPI Jumps 1.8% In Nov	-0.6	0.000	-0.011	0.000	-
Fed's Kaplan: Texas Unemployment Lowest Since '70s	0.82	-0.0272	0.009	0.000	+
DATA SNAP: US Jobless Rate Rises For 1st Time Since Dec	-0.82	-0.025	-0.016	-0.002	-

The table shows the headline of three news articles scored by Ravenpack, along with corresponding sentiment values computed using three different lexicons. It shows next to Ravenpack three sentiment values computed using different lexicons: Loughran-McDonald, the psychological Harvard-IV dictionary, and Hu and Liu. ESS is normalized to a range between -1 and 1 for comparison with the other sentiment scores. The last column shows the sentiment sign according to a human sentiment classification. The full articles are provided in Appendix.

later, positive sentiment on interest rates is associated with falling interest rates and, hence, increasing asset prices. This aligns with the perspective of fixed-income investors, whose asset holdings appreciate following a decrease in interest rates.

3.3. Relation to macroeconomic data

In this section, we examine the relationship between sentiment and macroeconomic variables. This analysis is particularly important, because variables such as inflation and unemployment are frequently employed to forecast interest rates. The first column of Fig. 2 shows the historical path of the standardized values of the macroeconomic variables and the sentiment data.³ There is a visible negative co-movement of interest rate sentiment and the change of interest rate, indicating that positive news is related to falling interest rates. For instance, in the pre-crisis period surrounding 2007, a rise in sentiment preceded decreasing interest rates. Similarly, in 2014, negative sentiment emerged, while the short rate remained in the zero-lower bound. The decrease in sentiment may have foreshadowed the eventual increase in rates that occurred one year later. Additionally, there appears to be a negative relationship between sentiment and the growth of the unemployment rate, which is particularly visible during the financial crisis and the coronavirus pandemic.

Next, we analyze whether sentiment has a predictive power over macroeconomic variables. The first two columns of Table 3 show the regression of inflation and unemployment rate on the three sentiment scores. The results suggest that interest rate and unemployment sentiment predict future unemployment rates. This evidence implies that sentiment has a predictive power over the real economy, potentially making it a useful indicator for interest rates movements. Later, we will demonstrate that this connection is linked to the relationship of sentiment and the slope of the yield curve.

³ We work with the 3-month Treasury yield extracted from the St. Louis Fed database. Personal Consumption Expenditures Excluding Food and Energy was used to compute annual inflation and was also downloaded from the St. Louis Fed database. Unemployment rate data were obtained from the Bureau of Labor Statistics database and transformed into yearly log-growth of unemployment rate.

3.4. Relation to expectations

In this section we examine the question of how sentiment and expectations are related. To proxy expectations, we use the three-month-ahead forecasts to expected median Treasury bill, CPI inflation, and unemployment rate by the Survey of Professional Forecasters. As the expectations data are available on a quarterly basis, the analysis is also conducted quarterly. Similar to the monthly sentiment, we aggregate the sentiment scores by their quarterly average.

Generally, we expect sentiment and the expectation measures to be related, since it is plausible to assume that forecasts are updated based on current news perceptions. However, we are interested on whether a measure of news perceptions has information which is not entailed in forecasts. To answer that we regress the three expectation measures on the three sentiment scores. The fifth column of Table 3 shows that more than 50% of variations in interest rates forecasts can be explained by sentiment. A one standard deviation higher sentiment score is associated with a decrease of 28 basis points in the interest rates expectations, *ceteris paribus*.

Additionally, the relation between sentiment and expectations is also illustrated in Fig. 2. The second column of the figure plots the sentiment scores and the expectations with the same underlying variables. The three-month-ahead forecast follows a similar path as the interest rate itself, and is thus negatively correlated with the interest rate sentiment. It falls abruptly in the beginning of the 2000s and during the great recession. A visual conclusion with respect to inflation and unemployment measures is more difficult, as the series are more noisy, although it is visible that both unemployment rate measures trends in opposite directions in the post-crisis period.

Finally, a crucial question is whether sentiment has predictive power over the macroeconomic variables beyond the measures of expectations. To answer this question, we regress inflation and unemployment rate on their sentiment and expectations together. The last two columns of Table 3 illustrate the results. Unemployment sentiment is better than expected unemployment in predicting true unemployment, but inflation sentiment is worse than expected inflation in predicting true inflation. This shows that sentiment and expectations indeed entail different information, both of which might be valuable to predict future macroeconomic variables.

An additional interpretation of these results concerns the rationality of economic agents when forming their expectations. If we interpret



Fig. 2. Historical relationships between economic variables and sentiment scores.

This figure shows different variables related to interest rates, inflation, and unemployment in blue superimposed by our sentiment measures in yellow. All time series are standardized. The left column displays the historical relationship between changes in interest rates, inflation rate, and unemployment growth, and their corresponding sentiment scores. The middle column shows the historical evolution of expectations of these three variables, while the right column shows the yield curve factors that are expected to be associated with them. The data cover the period from January 2000 to May 2020. For illustrative purposes we smoothed the sentiment scores using a Kalman smoother.

Table 3
Regression results of sentiment, expectations, and economic variables.

	(1) π	(2) u	(3) π^e	(4) u^e	(5) i^e	(6) π	(7) u
$L(S_t)$	-0.03 (0.02)	7.56*** (1.67)	-0.02 (0.30)	0.32 (0.24)	-0.28*** (0.04)		
$L(S_\pi)$	-0.01 (0.02)	1.00 (1.09)	0.11 (0.22)	-0.10 (0.21)	0.04 (0.03)	0.03 (0.03)	
$L(S_u)$	-0.003 (0.02)	-5.82*** (1.09)	-0.24 (0.20)	-0.14 (0.26)	0.002 (0.03)		-5.16* (2.97)
$L(\pi^e)$						0.06*** (0.02)	
$L(u^e)$							-1.45 (1.30)
Constant	1.71*** (0.02)	0.02 (1.05)	2.16*** (0.27)	5.88*** (0.22)	-0.06* (0.03)	1.56*** (0.04)	9.56 (8.91)
Observations	243	243	81	81	81	81	81
Adjusted R ²	-0.01	0.28	-0.03	0.01	0.52	0.14	0.04

The table shows OLS regression results on sentiment using different specifications. Columns (1) and (2) present the regression of inflation and unemployment rate on the three lagged sentiment scores. Columns (3) to (5) conduct the same regression but with expected inflation, unemployment rate, interest rates as dependent variables. Lastly, columns (6) and (7) show the regression results of inflation and unemployment rate against their respective lagged sentiment and expected values. Heteroskedastic-consistent standard errors are shown in parentheses. Sentiment scores are standardized. ***, ** and * denote significance at the 1%, 5%, and 10% level, respectively.

sentiment, analogous to Keynes (1936), as a source of agents irrationality, our findings provide evidence that such irrationality is important in predicting the future state of the economy, which goes beyond rational expectations. As Table 3 shows, this irrational component seems to be more relevant to predict future unemployment rate than future inflation. However, if expectations are formed based on current news perceptions they might not be rational. This is particular evident in the case of interest rates, since sentiment and expectations are strongly correlated. Overall our findings provide substantial evidence of an existing link between sentiment, expectations, and economic variables.

In the following section, we will see that sentiment explains interest rates movements beyond this link.

3.5. Relation to the yield curve

We use yield curve factors made available on the Fed homepage by Gürkaynak et al. (2007) and convert them to yields going from 3-, 12-, 24-, 36-, 48-, 60-, 72-, 84-, 96-, 108- and 120-months maturities. As our sample covers the period from 2000 to 2020, it is important to address the features of the yield curve during this time frame.

Relative to last century, the new period is characterized by lower yields, with an average yields level mostly below 6%. The Fed's response to shocks in the economy seems to be more aggressive than in the last century, as evidenced by the series of steps to ease in the periods 2000–2001, 2007–2008 and 2019–2020. Furthermore, forward guidance had played a much more significant role than in previous years, primarily because the Fed introduced it as a communication instrument in 2003, but also because of the zero lower bound (see, for example, Eijffinger et al., 2017). Lastly, the yield curve inverted before all three recessions recorded in this period, with the curve tending to steepen again during and after the recessions. These movements are well captured by the Nelson–Siegel family of models.

During the period between 2008 and 2015, the fed policy rate was constrained by the zero lower bound, which raises the question of whether our analysis might be impacted negatively as a result. To address this, we begin our analysis of the yield curve with the dynamic Nelson–Siegel Model (DNS), which allows sentiment to influence the yield curve through various factors, including long-term yields, the slope, and the curvature of the curve. However, not accounting for the zero lower bound could affect our short rate based term structure model. In this case, the estimated impact of sentiment on the yield curve might be weaker than the true effects.⁴

Economic theory suggests that a link between the level factor of the yield curve and expected inflation should exist (see, for example, Zhu and Rahman (2015)). Therefore, we investigate whether there is a relation between the level factor and inflation sentiment. Likewise, we compare the slope of the yield curve with unemployment rate sentiment, and the curvature with the interest rate sentiment. The third column of Fig. 2 illustrates the three yield curve factors derived from the DNS model, along with the sentiment scores. Our results suggest a negative co-movement between interest rate sentiment and the curvature factor, with a correlation of approximately -0.38 . Moreover, we find correlations of -0.45 between inflation sentiment and the level factor, as well as -0.22 between unemployment sentiment and the slope factor. These findings align with previous studies that have identified similar relationships between the underlying macroeconomic variables and the yield curve factors (see, for example, Zhu & Rahman, 2015).

4. Short rate models

4.1. Linear model

As we use a short rate based yield curve model, we begin our analysis with the relation between the short-term interest rate and sentiment. Our starting point is the Taylor rule, which specifies that the short rate reacts to two key variables: inflation and unemployment rate. The work by Clarida et al. (2000) extends the Taylor rule in a forward-looking fashion, allowing for expectations of future inflation and unemployment rate. As pointed out by Ang and Piazzesi (2003), this implies that any variable that forecasts inflation or real activity will enter the right-hand side of the equation. Commonly, a lagged nominal interest rate is included, which represents the gradual fashion in which central banks adjust interest rates, as well as an intercept which entails the unidentified real rate target and the inflation target (see, for example, Clarida et al., 2000).

As demonstrated by Table 3, sentiment exhibits a predictive power over unemployment and interest rates forecasts, implying that sentiment scores can assist in predicting interest rates. However, our objective is to understand whether sentiment has a predictive power beyond these variables, which is why we incorporate them as covariates. Our baseline model is formulated as follows:

$$i_{t+1} = \beta_0 + \beta_1 i_t + \beta_2 \pi_t + \beta_3 u_t + \beta_4 S_{i,t} + \beta_5 S_{\pi,t} + \beta_6 S_{u,t} + \beta_7 \gamma_t + \epsilon_{t+1}. \quad (1)$$

⁴ We leave models addressing the ZLB explicitly for future research. A possible candidate to extend our model would be the shadow-rate model in the manner of Christensen and Rudebusch (2015).

i_t denotes the interest rate, π_t the inflation rate, u_t the unemployment rate, and S_t the sentiment on each macroeconomic variable. γ denotes additional covariates, such as the forecast of short-term interest rates and the Michigan Consumer Sentiment Index. The latter is added as a robustness check, since the analysis of the expectations may lose valuable information due to the data being quarterly available. The University of Michigan Consumer Sentiment Index is meant to control for expectations while maintaining the monthly frequency.

Table 4 shows the main results of the short rate regression. We find a significant effect of interest rate sentiment on future short-term rate, which is maintained over the different regression specifications. According to the estimated results in our baseline regression, an increase of one standard deviation in the interest rate sentiment results in an eight basis point decrease of the short term interest rate, ceteris paribus. Since the coefficients of inflation and unemployment rate sentiment are not significantly different from zero, a third regression is run, which leaves these variables aside. The adjusted R^2 stays constant, whereas the information criteria decrease. Hence, restricting the regression to account solely for interest rate sentiment results in a better and more parsimonious model. The expectations regressions, which substitutes inflation and unemployment rate by their expectations and adds the expected change in interest rates shows that the negative in-sample predictive ability of sentiment interest rates goes beyond expectations. Our findings are also robust towards using the Michigan Consumer Sentiment Index.

Unlike previous studies, we have found a negative relationship between sentiment and the short-term interest rates. We hypothesize that this difference is due to our sentiment variable being based on the perception of news on interest rates only. As declining interest rates tend to improve the performance of fixed income securities, news reporting falling interest rates could be interpreted as positive by investors. In contrast, Hubert and Labondance (2017) uses sentiment from central bank statements which focus on current and future economic conditions. Therefore, if the central bank has a positive view of the current and future economic conditions, investors will expect an increase in interest rates. Likewise, Gotthelf and Uhl (2019) uses sentiment from news articles related to political, debt, and monetary topics, which are closely linked to the perception of economic conditions.

The fact that the autoregressive coefficient ρ , shown in Table 4, is very close to unit is noteworthy. As a robustness check we re-estimated the model using the first differences of the short rate (see Table C.11 in Appendix) and found no significant deviations from our results.

Since Table 4 demonstrates the predictive power of sentiment beyond the expectations, we simplify our further empirical results and do not add expectations. To check the robustness of our findings we recreated our main empirical results controlling for Consumer Sentiment Index and found no meaningful deviation from the results presented in the next sections.⁵

4.2. Regime-switching model

Several papers have recognized the problem of the unit root or near unit roots in the short rate process. This is a big issue that needs to be taken into account, since it can break down the assumption of stationarity of the short rate process (see, for example, Hamilton, 1988). As a possible solution, some studies proposed relating the unit root problem to a structural break by assuming, for instance, the Markov-Switching model, the TAR or some variation thereof. A few of these works are listed in Audrino (2006).

We assume that sentiment on interest rates determines the regimes in a threshold autoregressive model. Specifically, the parameters of Eq. (1) vary according to the interest rate sentiment score. To estimate

⁵ Results can be obtained from the authors upon request.

Table 4
Results of short rate regression on sentiment.

	Baseline Regression			Expectations Regression		Michigan Consumer Sentiment	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
i	0.981*** (0.010)	0.984*** (0.009)	0.983*** (0.009)	0.961*** (0.018)	0.973*** (0.015)	0.983*** (0.010)	0.983*** (0.010)
π	0.020 (0.045)	0.024 (0.043)	0.024 (0.042)			0.025 (0.041)	0.024 (0.041)
u	−0.002** (0.001)	0.000 (0.001)	−0.000 (0.001)			0.000 (0.001)	0.000 (0.0005)
π^e				0.004 (0.011)	0.002 (0.011)		
u^e				−0.012 (0.010)	−0.004 (0.008)		
S_i		−0.082*** (0.014)	−0.083*** (0.013)	−0.056** (0.027)	−0.053* (0.028)	−0.082*** (0.015)	−0.083*** (0.015)
S_π		0.007 (0.011)		−0.010 (0.025)		0.007 (0.011)	
S_u		0.002 (0.016)		−0.032* (0.018)		0.002 (0.016)	
i^e				0.080 (0.101)	0.042 (0.084)		
MCS						0.001 (0.016)	0.001 (0.015)
Constant	−0.026 (0.064)	−0.037 (0.060)	−0.036 (0.060)	0.119 (0.085)	0.053 (0.066)	−0.038 (0.057)	−0.037 (0.057)
AIC	−127.6	−169.6	−173.2	−47.2	−49	−167.6	−171.2
BIC	−110.1	−141.6	−152.2	−25.7	−32.3	−136.1	−146.7
Observations	244	244	244	81	81	244	244
Adjusted R ²	0.989	0.991	0.991	0.990	0.990	0.991	0.991

The first three columns show the regression of the U.S. 3-month Treasury yield on different variables. Column (1) uses inflation and unemployment rate, column (2) adds the three sentiment scores, and column (3) omits inflation and unemployment rate sentiment. Columns (4) and (5) use the same specification but replace inflation and unemployment rate with their expectations. Lastly, columns (6) and (7) control for the Michigan Consumer Sentiment Index. Heteroskedastic-consistent standard errors are shown in parentheses. Sentiment scores are standardized. ***, ** and * denote significance at the 1%, 5%, and 10% level, respectively.

the optimal threshold we follow [Enders \(2015\)](#) and conduct a grid-search while excluding the highest and lowest 15% of the values in the sample to ensure that the sample size is not unreasonably small.

[Table 5](#) presents the results of the estimated TAR model. Notably, the information criteria support the two-regimes model. Moreover, the estimated threshold is substantially larger than the division between positive and negative scores, with only 17% of the data falling above the threshold.⁶ In the first regime, sentiment effects are not statistically significant, while in the second regime, a one standard deviation increase in interest rates sentiment decreases the future short-term rate by 6.8 basis points, *ceteris paribus*. Lastly, as a robustness check we estimate the model in first differences and, as in the 1-regime case, no major variations are observed (see [Table C.12](#) in Appendix).

[Fig. 1](#) illustrates the close relationship between the estimated regimes and business cycles, with short-term interest rates switching to the high sentiment regime during periods of declining interest rates and economic downturns. Additionally, [Fig. B.10](#) in Appendix demonstrates that, unlike the regimes generated by sentiment, regimes generated by inflation, unemployment rate, and even the lagged short rate level, have no links to changes in interest rate trends or business cycles.

In conclusion, the predictive ability of interest rate sentiment over short-term interest rates can be leveraged to anticipate economic downturns, a capability that is unique compared to other variables.

5. Term structure models

5.1. Dynamic Nelson–Siegel model

To analyze the impact of sentiment on the yield curve, we use two different yield curve models. The first model is the Dynamic Nelson–Siegel Model, which is, according to [Diebold and Rudebusch \(2013\)](#), a

⁶ Since the mean of the interest rate sentiment is slightly above 50, the division between positive and negative under the standardized sentiment measure should be slightly below 0.

commonly used model among financial market practitioners and central banks, including the Board of Governors of the U.S. Federal Reserve System and the ECB. This model is favored because the constraints it imposes on the yields and prices comply with economic theory and because it provides a parsimonious approximation as well as a good smoothness.

The state space representation of the Dynamic Nelson–Siegel model begins with the measurement equation, where $y_t(\tau)$ denotes the yield at time t with maturity τ .⁷

$$\begin{pmatrix} y_t(\tau_1) \\ y_t(\tau_2) \\ \vdots \\ y_t(\tau_N) \end{pmatrix} = \begin{pmatrix} 1 & \frac{1-e^{-\tau_1\gamma}}{\tau_1\gamma} & \frac{1-e^{-\tau_1\gamma}}{\tau_1\gamma} - e^{-\tau_1\gamma} \\ 1 & \frac{1-e^{-\tau_2\gamma}}{\tau_2\gamma} & \frac{1-e^{-\tau_2\gamma}}{\tau_2\gamma} - e^{-\tau_2\gamma} \\ \vdots & \vdots & \vdots \\ 1 & \frac{1-e^{-\tau_N\gamma}}{\tau_N\gamma} & \frac{1-e^{-\tau_N\gamma}}{\tau_N\gamma} - e^{-\tau_N\gamma} \end{pmatrix} \begin{pmatrix} L_t \\ S_t \\ C_t \end{pmatrix} + \begin{pmatrix} \epsilon_t(\tau_1) \\ \epsilon_t(\tau_2) \\ \vdots \\ \epsilon_t(\tau_N) \end{pmatrix}, \quad (2)$$

where $\epsilon(\tau)$ is assumed to be an idiosyncratic factor and is thus uncorrelated to the other error terms. The transition equation, is a first-order vector autoregressive model:

$$X_t - \mu = \Phi(X_{t-1} - \mu) + \eta_t, \quad (3)$$

where the variables are

$$X_t = \begin{pmatrix} L_t \\ S_t \\ C_t \end{pmatrix}, \eta_t = \begin{pmatrix} \eta_t^L \\ \eta_t^S \\ \eta_t^C \end{pmatrix},$$

and the parameters vectors and matrices are

$$\mu_t = \begin{pmatrix} \mu^L \\ \mu^S \\ \mu^C \end{pmatrix}, \Phi = \begin{pmatrix} \phi_{11} & \phi_{12} & \phi_{13} \\ \phi_{21} & \phi_{22} & \phi_{23} \\ \phi_{31} & \phi_{32} & \phi_{33} \end{pmatrix}.$$

Further, we define the error terms from the transition and measurement equations as white noise processes that are orthogonal to each

⁷ See Appendix for details on the estimation.

Table 5

Results for sentiment-based regime-switching model.

	Constant	$lag(i)$	π	u	S_i	S_π	S_u	threshold
Regime 1	−0.111	0.911***	−0.030	0.003	0.091	−0.044	0.000	0.656
$S_{i,t} > c$	(0.344)	(0.039)	(0.202)	(0.006)	(0.082)	(0.059)	(0.068)	(0.438)
Regime 2	−0.038	0.997***	0.020	−0.000	−0.068***	0.004	0.013	
$S_{i,t} \leq c$	(0.054)	(0.010)	(0.038)	(0.001)	(0.013)	(0.009)	(0.017)	
Observations	Adjusted R ²	AIC	BIC					
244	0.992	−199.08	−146.62					

The table shows the results of the regime-switching model based on the threshold auto-regressive model. The threshold is based on interest rate sentiment. The first row shows the coefficients of the regime 1, which represents the high-sentiment regime. The second row presents the coefficients of regime 2, which stands for the low-sentiment regime. Heteroskedastic-consistent standard errors are shown in parentheses. The standard error for the threshold coefficient is estimated using blockwise bootstrapping. Sentiment scores are standardized. ***, ** and * denote significance at the 1%, 5%, and 10% level, respectively.

Table 6

Parameter estimates for dynamic Nelson–Siegel model.

	Estimated Φ -Matrix					
	L_t	S_t	C_t	$S_{i,t}$	$S_{\pi,t}$	$S_{u,t}$
μ	0.159 (0.099)	−0.145 (0.107)	−0.076 (0.244)	−0.318* (0.174)	0.361 (0.257)	1.275*** (0.227)
L_{t-1}	0.972*** (0.023)	0.021 (0.024)	−0.042 (0.056)	−0.079** (0.040)	−0.072 (0.059)	−0.315*** (0.052)
S_{t-1}	0.001 (0.018)	0.958*** (0.019)	0.031 (0.044)	0.140*** (0.032)	0.040 (0.047)	−0.017 (0.041)
C_{t-1}	0.014 (0.015)	0.025 (0.016)	0.886*** (0.037)	−0.115*** (0.027)	−0.028 (0.039)	−0.054 (0.035)
$S_{i,t-1}$	0.011 (0.027)	−0.083*** (0.029)	−0.035 (0.067)	0.638*** (0.048)	−0.137* (0.070)	−0.154** (0.062)
$S_{\pi,t-1}$	−0.025 (0.024)	0.029 (0.026)	−0.029 (0.060)	−0.006 (0.043)	0.201*** (0.063)	−0.032 (0.056)
$S_{u,t-1}$	−0.029 (0.029)	0.049 (0.032)	0.019 (0.073)	−0.014 (0.052)	0.069 (0.076)	0.155** (0.067)
	Estimated Correlation Matrix					
	L_t	S_t	C_t	$S_{i,t}$	$S_{\pi,t}$	$S_{u,t}$
L_t	1.000	−	−	−	−	−
S_t	−0.874***	1.000	−	−	−	−
C_t	−0.423***	0.253***	1.000	−	−	−
$S_{i,t}$	0.049	−0.207***	−0.152**	1.000	−	−
$S_{\pi,t}$	0.016	−0.026	−0.038	0.060	1.000	−
$S_{u,t}$	−0.062	0.065	0.037	0.009	−0.001	1.000
Obs.	γ	$RMSE_{3m}$	$RMSE_{2y}$	$RMSE_{5y}$	$RMSE_{8y}$	$RMSE_{10y}$
244	0.041	17.00	22.95	25.50	25.95	26.88

The table is divided into three parts. The upper table presents the coefficients of the first-order Vector Autoregression (VAR) process for yield curve factors and sentiment from January 2000 to May 2020. The yield curve factors are estimated using the Dynamic Nelson–Siegel model. Standard errors are shown in parentheses. The middle table reports the residual correlation matrix. The bottom table displays the shaping parameter estimates along with in-sample Root Mean Square Errors for different yields. ***, ** and * denote significance at the 1%, 5%, and 10% level, respectively.

other and to the initial states:

$$\begin{pmatrix} \eta_t \\ \epsilon_t \end{pmatrix} \sim WN \left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} Q & 0 \\ 0 & H \end{pmatrix} \right] \quad (4)$$

$$E(X_0 \eta_t') = 0$$

$$E(X_0 \epsilon_t') = 0 \quad (5)$$

Additionally, we assume that the H matrix is diagonal and that the Q matrix is non-diagonal. To extend the model to account for sentiment scores, we include them in a bidirectional causal fashion with the yield curve factors, such that $X_t' = (L_t, S_t, C_t, S_{i,t}, S_{\pi,t}, S_{u,t})$ and the dimensions of Φ , μ , η_t and Q are increased as appropriate. This setting allows sentiment to interact with the factors without spanning the yield curve.⁸

Table 6 shows the parameter estimates obtained from the DNS model. Our findings align with prior research, as we observe that the level, slope, and curvature are all highly persistent. Specifically, the

autoregressive coefficients for the three factors are 0.972, 0.958, and 0.886, respectively. The higher persistence of the level factor relative to the slope factor is consistent with the stylized fact that the yield dynamics are more persistent than spreads. However, the DNS model counterfactually implies that long-term yields are more persistent than short-term yields. This is due to the fact that long-term yields are primarily influenced by the level factor, whereas yields with shorter maturities are more closely tied to the slope factor.

Finally, with respect to the interaction between sentiment and the yield curve factors, we observe that positive lagged realizations of interest rate sentiment induce negative realizations of the (negative) slope. It is worth noting that the estimated coefficient, −0.083, is similar to the interest rate sentiment coefficient of the short rate model in Table 4. This similarity can be attributed to the fact that the slope is a linear combination of the long- and short-term yields. As we control for the long-term yields using the level factor, we obtain a significant effect of sentiment on the slope with a similar magnitude. Moreover, our findings indicate that sentiment does not have a significant impact on the long-term yields.

⁸ The question whether sentiment scores span the yield curve is beyond the scope of this paper. For simplicity, we assume they do not. See, for example, Bauer and Hamilton (2018) for implications about this assumption.

5.2. Arbitrage-free short rate based term structure model

Based on our previous findings, we conclude that sentiment impacts the short end of the yield curve only. Therefore, a yield curve model that integrates sentiment through short rate dynamics within an arbitrage-free framework may better capture the true effects of sentiment in the yield curve. In order to maintain consistency with our previous framework, we opt to use the Arbitrage-Free Short Rate based Term Structure Model from Nyholm (2018), which is also a member of the Nelson–Siegel family.

Next, we introduce the AFSR model in discrete time.⁹ Let X_t^f be a vector containing the four latent factors and X_t^s be the vector of sentiment scores. The short rate follows an affine form of state variables:

$$i_t = \rho_0 + (\rho_1^f)' X_t^f + (\rho_1^s)' X_t^s. \quad (6)$$

where $\rho_1^f = (1 \ 0 \ 0 \ 0)'$ and $\rho_1^s = 0$

Under the risk-neutral measure $\mathbb{Q}$, the state variables follow a first-order VAR process:

$$\begin{pmatrix} X_t^f \\ X_t^s \end{pmatrix} = \mu^{\mathbb{Q}} + \begin{pmatrix} \Phi_{f,f}^{\mathbb{Q}} & 0 \\ \Phi_{f,s}^{\mathbb{Q}} & \Phi_{s,s}^{\mathbb{Q}} \end{pmatrix} \begin{pmatrix} X_{t-1}^f \\ X_{t-1}^s \end{pmatrix} + \epsilon_t^{\mathbb{Q}}, \quad (7)$$

where the transition matrix for the unobservable variables is equal to:

$$\Phi_f^{\mathbb{Q}} = \begin{pmatrix} 1 & 1-\gamma & 1-\gamma & 1-\gamma \\ 0 & \gamma & \gamma-1 & \gamma-1 \\ 0 & 0 & \gamma & \gamma-1 \\ 0 & 0 & 0 & \gamma \end{pmatrix}. \quad (8)$$

The price of the zero-coupon bond is:

$$P_{t,\tau} = \exp(A_\tau + B_\tau' X_t^f). \quad (9)$$

where

$$A_\tau = -\rho_0 + A_{\tau-1} + B_{\tau-1}' \mu^{\mathbb{Q}} + \frac{1}{2} B_{\tau-1}' \Omega_f B_{\tau-1} \quad (10)$$

$$B_\tau^f = - \left[\sum_{k=0}^{\tau-1} \left(\Phi_f^{\mathbb{Q}} \right)^k \right]' \cdot \rho_1^f. \quad (11)$$

To simplify the notation, the variance–covariance matrix is defined as $\Sigma \Sigma' = \Omega$. Therefore, we find the adjustment term A by letting Ω_f be the upper left block of the variance–covariance matrix corresponding to the factor innovations in the state dynamics.

Based on Nyholm (2018), we include a monetary policy rule in the transition matrix under the $\mathbb{P}$ -measure:

$$(\Phi^{\mathbb{P}})' = \begin{pmatrix} \phi_{(1,1)} & \phi_{(1,2)} & \phi_{(1,3)} & \phi_{(1,4)} & 0 & 0 & 0 \\ 0 & \phi_{(2,2)} & \phi_{(2,3)} & \phi_{(2,4)} & 0 & 0 & 0 \\ 0 & \phi_{(3,2)} & \phi_{(3,3)} & \phi_{(3,4)} & 0 & 0 & 0 \\ 0 & \phi_{(4,2)} & \phi_{(4,3)} & \phi_{(4,4)} & 0 & 0 & 0 \\ \phi_{(5,1)} & 0 & 0 & 0 & \phi_{(5,5)} & 0 & 0 \\ \phi_{(6,1)} & 0 & 0 & 0 & 0 & \phi_{(6,6)} & 0 \\ \phi_{(7,1)} & 0 & 0 & 0 & 0 & 0 & \phi_{(7,7)} \end{pmatrix} \quad (12)$$

The first column of the above matrix $(\Phi^{\mathbb{P}})'$ is the short rate channel through which sentiment interacts with the yield curve. Columns 2 to 4 allow for interactions among the yield curve factors, while columns 5 to 9 represent the autoregressive processes for sentiment.

A big challenge in the application of no-arbitrage models concerns estimation and inference. According to Li et al. (2011), even though Maximum Likelihood Estimation (MLE) methods are suitable for state-space models with latent variables, finding a global optimum with a high dimension of parameters, typical in multi-factor term structure models, can be challenging. Based on the work of Joslin et al. (2011), which shows that the maximum likelihood function can be partitioned

such that the $\mathbb{Q}$ - and $\mathbb{P}$ -measure parameters can be estimated separately, Nyholm (2018) developed a stepwise estimation method to estimate the AFSR, which is explained in detail in Appendix.

Table 7 shows the estimated $\Phi^{\mathbb{P}}$ -matrix, residuals correlation matrix, γ , $\mu_t^{\mathbb{Q}}$, and in-sample RMSE for different yields. Unlike the DNS model, the AFSR model correctly captures the observations that short-term yields are more persistent than long-term yields. This is because, in the AFSR model, the long-term yield is a linear combination of the slope and the short rate factors.

Overall, the short rate dynamics implied by the AFSR model are consistent with our previous results, as sentiment has a significantly negative impact on the short rate. Notably, the lagged short rate factor does not have an impact on the slope of the yield curve in this model, suggesting that sentiment has no lagged effects on the slope once the short rate dynamics are accounted for. However, the positive and significant correlation of the residuals imply that sentiment shocks and slope shocks are positively correlated. Furthermore, our findings suggest the existence of a contemporaneous correlation of interest rates sentiment shocks and the three yield curve factor shocks.

5.3. Impulse response functions

To further understand the relation between sentiment and the yield curve factors, we use the AFSR model to generate impulse response functions (IRF). We adopt following recursive order of variables: (1) Interest rate sentiment, (2) inflation sentiment, (3) unemployment sentiment, (4) short rate, (5) slope, (6) curvature 1 and (7) curvature 2. Our choice reflects our focus based on understanding the impact of sentiment shocks on yields. While this approach assumes a contemporaneous unidirectional effect from sentiment on yields, it does not deviate substantially from our model assumptions.

Fig. 3 shows the factor responses to sentiment shocks with confidence bounds plotted in red. Interest rates sentiment has a negative impact on the short rate factor, which accumulates for up to 10 periods before converging to zero. The slope reacts with a significant immediate increase of around 0.05 units, but the speed of decay is unclear, given the wide confidence bounds. The first curvature factor also exhibits a statistically significant decline of around 0.12 units. The effects from inflation and unemployment rate sentiment shocks are not significant.

Finally, the analysis using the DNS model, shown in Fig. B.11 in Appendix confirms that the response of the level to interest rate sentiment shock is not significant; however, the effects on the slope are significant over the first 20 periods.

In conclusion, our analysis suggests that interest rate sentiment has a long-lasting negative impact on the short end of the yield curve. Additionally, the effects of sentiment on the slope are positive, but short-lasting. These findings are in line with the observation that interest rates sentiment predicts high unemployment rates, and in the case of the regime-switching model, economic downturns, as the slope tends to rise during these periods.

5.4. Regime-switching yield curve model

To address regime changes, we integrate the TAR model in our yield curve models. As the DNS model does not explicitly account for the short rate, we assume that long-term yields are subject to sentiment-based regimes. Additionally, we assume that the dynamics of the slope and curvature factors are not subject to regime changes, although, they will be indirectly impacted by the regimes, since the parameters of the short-term and the long-term rates are.

Table 8 shows that with regard to the DNS model, the parameters do not significantly differ between the two regimes, except for a significant effect of the curvature on the level factor in the high sentiment regime. In the second regime, the level factor follows an AR(1) process. Our findings on the short rate closely resemble our previous results, as interest rate sentiment remains significant in the second regime, which is characterized by relatively low sentiment levels.

⁹ See Appendix for details on the derivation of the model.

Table 7

Parameter estimates for Arbitrage-free short rate based model.

Estimated Φ^P -Matrix							
	i_t	S_t	C_t^1	C_t^2	$S_{i,t}$	$S_{\pi,t}$	$S_{u,t}$
μ^P	-0.013 (0.018)	0.048 (0.084)	-0.198 (0.230)	0.177 (0.219)	0.005 (0.043)	-0.008 (0.062)	0.000 (0.058)
i_{t-1}	0.995*** (0.007)	-0.000 (0.020)	-0.014 (0.053)	-0.016 (0.051)	—	—	—
S_{t-1}	—	0.951*** (0.022)	-0.055 (0.063)	-0.001 (0.059)	—	—	—
C_{t-1}^1	—	-0.035* (0.016)	0.913*** (0.045)	0.005 (0.043)	—	—	—
C_{t-1}^2	—	-0.037* (0.018)	0.012 (0.049)	0.907*** (0.047)	—	—	—
$S_{i,t-1}$	-0.092*** (0.012)	—	—	—	0.733*** (0.042)	—	—
$S_{\pi,t-1}$	-0.006 (0.011)	—	—	—	—	0.255*** (0.062)	—
$S_{u,t-1}$	0.005 (0.012)	—	—	—	—	—	0.409*** (0.059)
Estimated Correlation Matrix							
	i_t	S_t	C_t^1	C_t^2	$S_{i,t}$	$S_{\pi,t}$	$S_{g,t}$
i_t	1.000	—	—	—	—	—	—
S_t	-0.481***	1.000	—	—	—	—	—
C_t^1	0.043	-0.396***	1.000	—	—	—	—
C_t^2	-0.298***	0.399***	-0.671***	1.000	—	—	—
$S_{i,t}$	-0.374***	0.173***	-0.127**	0.017	1.000	—	—
$S_{\pi,t}$	-0.001	-0.063	0.050	-0.095	0.054	1.000	—
$S_{g,t}$	-0.001	-0.078	0.064	-0.038	-0.005	0.062	1.000
Obs.	γ	μ_t^Q	$RMS E_{3m}$	$RMS E_{2y}$	$RMS E_{5y}$	$RMS E_{8y}$	$RMS E_{10y}$
244	0.951	2.22	18.91	23.00	26.60	27.08	27.93

The table is divided into three parts. The upper table presents the coefficients of the first-order Vector Autoregression (VAR) process for yield curve factors and sentiment from January 2000 to May 2020. The yield curve factors are estimated using the Arbitrage-free Short Rate based Model. Standard errors are shown in parentheses. The middle table reports the residual correlation matrix. The bottom table displays the shaping parameter estimates along with in-sample Root Mean Square Errors for different yields. ***, ** and * denote significance at the 1%, 5%, and 10% level, respectively.

Table 8

Parameter estimates for Regime-switching yield curve models.

Panel A: Level Factor from Dynamic Nelson–Siegel Model								
	μ	L_{t-1}	S_{t-1}	C_{t-1}	$S_{i,t-1}$	$S_{\pi,t-1}$	$S_{u,t-1}$	threshold
Regime 1	0.332	0.996***	−0.035	0.109**	−0.057	−0.148	0.018	0.403
$S_{i,t} > c$	(0.271)	(0.044)	(0.032)	(0.050)	(0.075)	(0.107)	(0.068)	(0.296)
Regime 2	0.172	0.971***	0.017	−0.010	0.040	0.014	−0.053	
$S_{i,t} \leq c$	(0.106)	(0.022)	(0.016)	(0.019)	(0.036)	(0.033)	(0.049)	
Observations	Adjusted R ²	AIC	BIC					
244	0.944	195.6	248.0					
Panel B: Short Rate Factor from Arbitrage-free Short Rate based Model								
	μ	i_{t-1}	S_{t-1}	C_{t-1}	$S_{i,t-1}$	$S_{\pi,t-1}$	$S_{u,t-1}$	threshold
Regime 1	−0.306	0.895***	−	−	0.179	−0.084	0.039	0.887
$S_{i,t} > c$	(0.221)	(0.045)			(0.108)	(0.056)	(0.059)	(0.489)
Regime 2	−0.002	1.007***	−	−	−0.058***	0.003	0.009	
$S_{i,t} \leq c$	(0.013)	(0.006)			(0.013)	(0.010)	(0.010)	
Observations	Adjusted R ²	AIC	BIC					
244	0.989	−118.6	−80.17					

Panel A presents the regression results for the threshold autoregressive Dynamic Nelson–Siegel model. Panel B shows the results for the threshold autoregressive Arbitrage-Free Short Rate based model. The former includes a sentiment-based threshold in the level process, while the latter includes it in the short rate process. Standard errors are shown in parentheses. The standard error for the threshold coefficient is estimated using blockwise bootstrapping. Sentiment scores are standardized. ***, ** and * denote significance at the 1%, 5%, and 10% level, respectively.

5.5. Forecasting

The ultimate test for sentiment is its ability to forecast the yield curve out-of-sample. In fact, a motivation for [Diebold and Li \(2006\)](#) to develop the DNS was that “interest rate point forecasting is crucial for bond portfolio management, and interest rate density forecasting is important for both derivatives pricing and risk management”. Still, as pointed out by [Diebold and Rudebusch \(2013\)](#), the DNS and every other yield curve model tend to perform worse than simply assuming that each factor follows a random walk.

We use the Root Mean Square Error (RMSE) to evaluate the performance of 13 different models. The first group of models contains three different variations of the Taylor rule¹⁰: (1) the Original Taylor rule, (2) the Sentiment Taylor rule, which makes use of the three sentiment scores, and (3) the Restricted Taylor rule, which only uses interest rate

¹⁰ We refer to the Taylor rule when we include inflation and unemployment rate as explanatory variables.

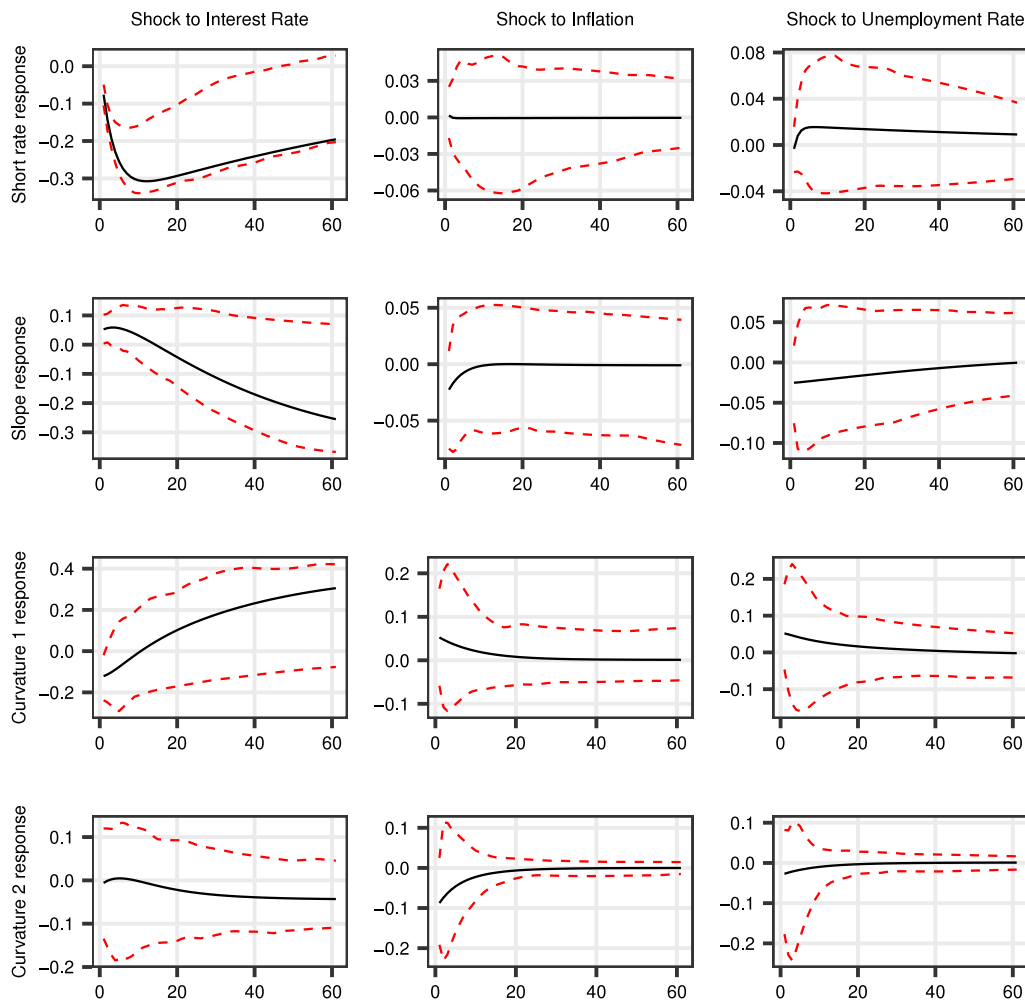


Fig. 3. Impulse response functions of yield curve factors to sentiment shocks.

The figure displays the impulse response functions of yield curve factors to sentiment shocks using the Arbitrage-free Short Rate based Model. The black line represents the impulse response path for 0 to 60 periods ahead, while the red dotted lines represent the 90% confidence intervals.

sentiment. Likewise, we include three variations of an AR(1) model¹¹: (1) the simple AR(1) process, (2) AR(1) including the three sentiment scores, and (3) AR(1) with interest rate sentiment. Each model assumes one and then two regimes for the short rate in the case of the AFSR and level in the case of the DNS. Lastly, we compare these models with a Random Walk (RW) without drift.

A Model Confidence Set (MCS) is performed to check for statistical significance, as we prefer not to assume a benchmark model to be compared against, but rather to create a confidence set based on the best model and to reject all other models which perform significantly worse (see, for example, Hansen et al., 2011). We run the forecast on a rolling basis starting at the beginning of 2012; i.e. the model is re-estimated each month, and the sample is expanded. Although the outcome of the tests might vary depending on the sample split, the ranking of the models with respect to RMSE is mostly robust toward sample splitting.¹²

Table 9 shows the DNS model's forecasting results for 1 to 12 months ahead, using yields with maturities of 3-months, 2-, 5-, 8-, and 10-years. The table highlights the minimum RMSE for each forecast category, displayed in bold text. Our findings indicate that the random

walk model overperforms all model variations with and without sentiment in almost all categories. The TAR model employing only interest rate sentiment exhibits the most accurate predictions for the 12-month forecasting horizon when applied to the 3-month yield. Nevertheless, when the model confidence set is applied, the overperformance of the random walk is rarely significant.

The AFSR model demonstrates that the use of sentiment can enhance yield forecasting accuracy, as Table 10 shows. Among the various models, two consistently exhibit the lowest RMSE across the curve: the simple TAR(1) and the TAR(1) that includes interest rate sentiment. The simple TAR(1) is the most effective model for 1-month forecasting. This finding also supports the assumption that sentiment contains predictive information because, even though it does not influence the local regime dynamics, it contributes to the model by establishing the regimes.

For the 3-month yield, the performance of all models is not statistically different. However, for the other maturities, the results vary depending on the forecasting horizon. For instance, the Taylor rule models are eliminated from the 75% confidence set for 2- and 3-month forecasting horizons. The random walk is eliminated from 6-month forecasting of the 2-year yield and of the 2- and 3-month forecasting from the 8- and 10-year yields. Overall, the model confidence set confirms that the 2-regime models perform as well as, or better than, the other models.

We finish this section with a discussion of the forecasting accuracy of the two Nelson-Siegel models. Although the aim of the paper is

¹¹ We refer to this group as AR(1), because the factors follow an AR(1) process when sentiment is not included.

¹² See Tables C.14 and C.15 in Appendix for the forecasting results with a rolling window starting in 2016.

Table 9

Out-of-sample forecast using sentiment in the dynamic Nelson–Siegel model.

	1-Month		2-Months		3-Months		6-Months		12-Months	
	1 Regime	2 Regimes	1 Regime	2 Regimes	1 Regime	2 Regimes	1 Regime	2 Regimes	1 Regime	2 Regimes
3-Months Yield										
Taylor Rule										
(1)	17.86**	33.17**	27.84**	32.42**	37.11**	37.86**	56.84**	48.07**	91.52*	69.67**
(2)	16.59**	31.58	26.35**	33.49**	35.08**	39.70**	53.58**	49.81**	90.13**	70.46**
(3)	16.52**	32.01**	26.31**	31.73**	34.98**	37.13**	53.40**	46.28**	89.59**	68.55**
AR(1)										
(1)	18.01**	23.39**	29.71**	30.33**	39.76**	36.62**	60.16*	48.24**	84.15**	68.26**
(2)	16.43**	22.31**	27.01**	29.42**	36.61**	36.41**	56.18**	47.32**	85.44**	69.17**
(3)	16.38**	19.42**	26.95**	26.42**	36.51**	32.81**	56.04**	43.37**	84.40**	66.72**
Random Walk	15.82**		24.29**		31.29**		43.35**		67.92**	
2-Years Yield										
Taylor Rule										
(1)	16.85*	33.51*	26.27**	33.48**	35.58**	38.93**	56.24**	49.44**	87.36*	70.99**
(2)	16.14*	33.08	25.24**	35.04**	33.88**	40.31**	53.68**	51.29**	86.07*	71.06**
(3)	16.06**	33.02*	25.02**	33.52**	33.64**	38.57**	53.41**	48.48**	85.75*	70.19**
AR(1)										
(1)	16.96*	25.81*	26.87**	30.60**	36.52**	36.69**	57.68**	48.74**	85.18*	69.81**
(2)	16.25*	25.48	25.50**	30.76**	34.43**	36.81**	54.64**	48.86**	84.76*	69.94**
(3)	16.17*	23.31*	25.32**	28.18**	34.29**	33.98**	54.59**	45.50**	84.47*	68.93**
Random Walk	15.21**		22.24**		29.16**		43.36**		69.23**	
5-Years Yield										
Taylor Rule										
(1)	21.66**	34.91*	32.47**	37.94**	42.47**	43.86**	63.56**	54.72**	87.52*	74.43**
(2)	21.82**	35.14	32.12**	39.53**	41.73**	45.13**	62.05**	56.43**	86.93*	74.20**
(3)	21.40**	34.29*	31.66**	37.77**	41.28**	43.43**	61.71**	54.16**	86.86*	74.22**
AR(1)										
(1)	21.24**	28.90**	31.82**	34.88**	41.62**	41.20**	62.61**	53.35**	86.99*	73.20**
(2)	21.41**	29.62	31.62**	36.00*	41.06**	42.08**	61.12**	54.16**	86.40*	72.55**
(3)	20.98**	27.27**	31.25**	33.51**	40.75**	39.62**	61.02**	51.42**	86.66*	73.08**
Random Walk	19.32**		27.24**		33.94**		48.32**		72.26**	
8-Years Yield										
Taylor Rule										
(1)	23.52**	35.49**	35.62**	40.14**	46.35**	46.38**	68.18**	57.54**	92.57**	77.72**
(2)	24.18**	36.34	35.90**	41.98**	46.44**	47.97**	67.42**	59.28**	92.22**	77.36**
(3)	23.68**	35.34**	35.41**	40.32**	45.95**	46.36**	67.08**	57.46**	92.31**	77.93**
AR(1)										
(1)	22.65**	30.19**	34.42**	37.37**	44.74**	44.04**	66.14**	56.37**	92.63*	76.91**
(2)	23.50**	31.18	35.14**	38.59**	45.38**	45.03**	65.62**	57.04**	91.52**	75.38**
(3)	22.98**	29.25**	34.74**	36.67**	45.04**	43.21**	65.51**	55.14**	92.13*	76.96**
Random Walk	20.71**		29.83**		36.79**		50.99**		75.00**	
10-Years Yield										
Taylor Rule										
(1)	24.08*	36.16*	36.42**	41.41**	47.33**	47.73**	69.71**	59.32**	95.65**	80.13**
(2)	24.78*	37.17	36.86**	43.26**	47.69**	49.38**	69.22**	60.99**	95.30**	79.72**
(3)	24.32*	36.38*	36.41**	41.90**	47.22**	48.01**	68.88**	59.53**	95.46**	80.51**
AR(1)										
(1)	23.28**	31.19*	35.27**	38.88**	45.71**	45.66**	67.57**	58.39**	96.28*	79.64**
(2)	24.15*	31.92	36.13**	39.79**	46.59**	46.38**	67.18**	58.68**	94.60**	77.59**
(3)	23.67*	30.55*	35.77**	38.44**	46.29**	45.11**	67.09**	57.40**	95.38*	79.64**
Random Walk	22.07**		31.53**		38.56**		52.94**		77.28**	

The table presents the root mean square error (RMSE) of 1-, 2-, 3-, 6-, and 12-month ahead forecasts for the U.S. Treasury yield curve from January 2000 to May 2020 estimated with the Dynamic Nelson–Siegel Model. The models are re-estimated each month, with the sample size increasing by one observation point each time, starting in January 2012. The models with the lowest RMSE for a particular maturity are shown in bold. * and ** indicate the models in the 90% and 75% superior confidence set, respectively.

not to compare the two models, it is useful to highlight their main contrasts. On the one hand, since sentiment demonstrates an in-sample predictive power on the short rate, we expected sentiment to improve yield forecasting accuracy when incorporating the short rate channel in the term structure model. On the other hand, the constraints imposed in the AFSR model could put it at a disadvantage compared to the DNS model.

The AFSR model clearly improves the forecasting accuracy of 2-regime sentiment models. However, the results are mixed when comparing each 1-regime model with its equivalent in the DNS model except for the random walk, which performs better in the DNS model. Therefore, the restrictions imposed in the AFSR model do not necessarily harm the overall forecasting accuracy. In fact, our findings show that the TAR(1) within the AFSR framework produces the overall lowest RMSE for the short end of the yield curve, outperforming the

random walk in the DNS framework. We conclude that using a 2-regime sentiment based model can improve the overall forecasting accuracy of short-term yields.

6. Limitations

This section addresses different limitations of our study. One limitation lies in our models' inability to account for the heterogeneity of individuals and their varying interpretations of news articles. Based on our definition of sentiment, it is reasonable to assume that market participants may perceive news differently. In our approach, however, the sentiment scores reflect the collective perception of the majority of readers, effectively representing the sentiment of a representative agent. Therefore, our analysis operates under the simplifying assumption that only the sentiment of this representative agent is relevant, which makes our results more tractable.

Table 10

Out-of-sample forecast using sentiment in the arbitrage-free short rate based model.

	1-Month		2-Months		3-Months		6-Months		12-Months	
	1 Regime	2 Regimes	1 Regime	2 Regimes	1 Regime	2 Regimes	1 Regime	2 Regimes	1 Regime	2 Regimes
3-Months Yield										
Taylor Rule										
(1)	17.37**	16.11**	26.83**	24.40**	34.89**	31.07**	48.16**	41.16**	75.32**	65.95**
(2)	16.79**	17.15**	25.90**	24.52**	32.11**	31.07**	36.77**	41.20**	60.22**	64.66**
(3)	16.65**	16.54**	25.79**	23.98**	32.09**	30.45**	37.27**	40.60**	60.68**	64.10**
AR(1)										
(1)	15.97**	15.07**	24.98**	23.37**	32.18**	30.30**	44.36**	39.98**	68.59**	65.49**
(2)	16.79**	16.98**	25.76**	24.43**	32.17**	31.15**	37.37**	41.43**	59.68**	64.82**
(3)	16.48**	15.77**	25.43**	22.82**	31.82**	29.28**	37.47**	38.81**	59.18**	63.37**
Random Walk	15.99**		25.08**		32.38**		44.44**		68.05**	
2-Years Yield										
Taylor Rule										
(1)	19.73**	18.41**	30.96*	24.98**	41.10*	31.59**	58.51*	43.87**	84.73*	69.23**
(2)	18.57**	19.09**	29.16*	24.95**	36.81**	31.12**	48.43**	44.15**	71.39**	68.60**
(3)	18.41**	18.43**	29.06**	24.50**	36.85**	30.72**	48.96**	43.69**	71.83**	68.10**
AR(1)										
(1)	17.45**	16.76**	27.45**	23.78**	36.37**	30.60**	51.40*	42.80**	74.75**	68.85**
(2)	18.56**	19.28**	29.01*	25.12**	36.70**	31.38**	47.43**	44.36**	68.96**	68.67**
(3)	18.07**	17.36**	28.47**	23.57**	36.19**	29.71**	47.42**	42.28**	68.40**	67.61**
Random Walk	17.22**		27.28**		36.38**		51.60*		74.34**	
5-Years Yield										
Taylor Rule										
(1)	24.38**	23.33**	36.33*	30.57**	46.63*	37.00**	63.50**	48.69**	86.30**	71.61**
(2)	23.35**	23.90**	34.91*	30.78**	42.89**	36.83**	56.19**	49.52**	76.59**	72.09**
(3)	23.24**	23.38**	34.83**	30.40**	42.94**	36.52**	56.66**	49.20**	76.85**	71.71**
AR(1)										
(1)	22.34**	21.87**	32.95**	29.39**	41.91**	35.89**	56.16**	47.60**	77.60**	71.21**
(2)	23.38**	24.11**	34.89*	31.00**	42.90*	37.13**	55.76**	49.80**	76.82**	72.36**
(3)	22.98**	22.46**	34.38**	29.63**	42.44**	35.63**	55.74**	48.13**	76.22**	71.45**
Random Walk	22.16**		32.90**		41.98**		56.28**		75.42**	
8-Years Yield										
Taylor Rule										
(1)	26.04*	25.01**	39.97	33.56**	51.42*	40.44**	70.44**	52.87**	94.27**	75.84**
(2)	24.95**	25.49**	38.78*	33.76**	48.13*	40.34**	64.53**	53.73**	85.64**	76.62**
(3)	24.90**	25.05**	38.75*	33.45**	48.24*	40.06**	65.05**	53.51**	86.09**	76.35**
AR(1)										
(1)	24.15**	23.71**	36.57*	32.45**	46.36*	39.32**	61.91**	51.75**	82.63**	75.33**
(2)	24.97**	25.71**	38.74*	33.96**	48.12*	40.62**	64.27**	54.00**	86.52*	76.95**
(3)	24.68**	24.26**	38.35*	32.75**	47.76*	39.23**	64.34**	52.55**	86.18**	76.11**
Random Walk	23.75**		36.23*		46.09*		62.24**		82.09**	
10-Years Yield										
Taylor Rule										
(1)	25.52**	24.70**	39.99*	33.75**	51.93*	40.79**	72.38**	53.97**	97.32**	77.34**
(2)	24.43**	24.85**	39.12*	33.70**	49.14*	40.54**	67.44**	54.68**	89.41**	78.09**
(3)	24.46**	24.50**	39.15*	33.50**	49.28*	40.33**	67.97**	54.55**	89.91**	77.89**
AR(1)										
(1)	24.34**	23.86**	37.46*	33.00**	47.72*	39.98**	64.61**	53.06**	85.49**	76.88**
(2)	24.49**	25.07**	39.20*	33.90**	49.31*	40.84**	67.54**	54.96**	90.67	78.45**
(3)	24.41**	24.01**	38.98*	33.01**	49.09*	39.72**	67.70**	53.76**	90.46*	77.69**
Random Walk	24.10**		37.13*		47.32*		64.94**		85.96**	

The table presents the root mean square error (RMSE) of 1-, 2-, 3-, 6-, and 12-month ahead forecasts for the U.S. Treasury yield curve from January 2000 to May 2020 estimated with the Arbitrage-Free Short Rate based Model. The models are re-estimated each month, with the sample size increasing by one observation point each time, starting in January 2012. The models with the lowest RMSE for a particular maturity are shown in bold. * and ** indicate the models in the 90% and 75% superior confidence set, respectively.

We argue that this simplifying assumption makes our results, and especially our forecasting accuracy rather conservative, because noisy economic agents, that interpret news in a different way than the majority, should decrease the signal to noise ratio of our sentiment scores, impacting our forecasting accuracy negatively. Therefore, accounting for this heterogeneity should yield further improvements in our results.

Another limitation is the fact that our analysis is limited to the United States. While a cross-country analysis can offer valuable insights into various factors, we believe that our findings remain largely unaffected by specific U.S. characteristics. The behavior of yield curves across different developed countries exhibits substantial similarity in terms of key characteristics. For instance, yield curve inversions tends to predict recessions in other countries as well (see, for example, [Chinn & Kucko, 2015](#); [Sabes & Sahuc, 2023](#)).

Likewise, the effect of sentiment on the stock market remains consistent across various countries, as shown by [Schmeling \(2009\)](#). He

finds that sentiment predicts a decrease of stock market returns across 18 countries. Nevertheless, [Schmeling \(2009\)](#) also unveils that the strength of the impact of sentiment on the stock market is stronger for countries with less market integrity and with higher susceptibility to exhibit herd-like behavior and overreaction. Notably, these tendencies are typically more pronounced in developing countries. Therefore, our interpretation of the results grounded in U.S. data can be viewed as somewhat conservative and expanding our analysis to encompass other countries would likely amplify the statistical power of news sentiment.

7. Conclusion

We provide substantial new evidence of an existing effect of macroeconomic news sentiment on the yield curve. Specifically, we demonstrated that interest rates sentiment has an in-sample predictive power

over the short-term interest rate, which goes beyond any information conveyed by current state variables and their expectations. Our analysis reveals that positive perceptions of news articles related to interest rates are indicative of lower future short-term rates. Additionally, we found that sentiment can be used to identify two interest rates regimes, which correspond to different states of the economy. High sentiment regimes are associated with decreasing interest rates and beginning of economic recessions.

Lastly, we analyzed the term structure of U.S. bond yields assuming two different models: the Dynamic Nelson–Siegel model and the Arbitrage-free Short Rate based model. The latter builds on the former by incorporating arbitrage-free assumptions and a short rate channel. Using both models, we found that interest rates sentiment affects the short end and the slope of the yield curve. When news related to interest rates are relatively positive, short-term yields tend to fall relative to long-term yields, resulting in an increased slope. These findings are consistent with the observation that high interest rates sentiment predicts high unemployment rates. Our analysis also demonstrates the out-of-sample predictive power of sentiment. When sentiment is used to establish the regimes of the short rate, it leads to an improvement in accuracy of short-term yield forecasts, outperforming the random walk.

Our research carries significant implications for both policymakers and private market participants. From a policy perspective, our findings suggest that variables beyond expectations and macroeconomic variables help predicting movements in the yield curve. Central banks can consider incorporating sentiment into their forecasting models to enhance their understanding of the repercussions of their monetary policy choices on future outcomes as well as to better assess the future trajectory of the economy following shocks, so they can make more informed decisions regarding appropriate responses. For private investors, the accurate prediction provided by sentiment offers opportunities to develop more effective trading strategies that are grounded in the analysis of sentiment surrounding the yield curve. Understanding the effects of sentiment shocks on the yield curve also improves the understanding of abrupt market movements and helps better managing risks.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Appendix A. Theoretical appendix

A.1. Arbitrage-free short rate based model

Let the transition equation for the 4-dimensional vector of factors X_t be given by a first-order VAR process:

$$X_t = \mu^{\mathbb{P}} + \Phi^{\mathbb{P}} X_{t-1} + \Sigma \epsilon_t^{\mathbb{P}}, \epsilon_t^{\mathbb{P}} \sim N(0, I), \quad (\text{A.1})$$

where I is the identity matrix. The short rate equation is denoted by

$$i_t = \rho_0 + \rho_1' X_t, \quad (\text{A.2})$$

where ρ_0 is a scalar, and ρ_1 is a 4×1 vector.

The price of risk also takes an affine form on the state vector:

$$A_t = \lambda_0 + \lambda_1' X_t, \quad (\text{A.3})$$

where λ_0 is a 4×1 vector, and λ_1 is a 4×4 matrix.

Under no-arbitrage, the current bond price at time t with maturity τ is given by following equation:

$$P_{t,\tau} = E_t^{\mathbb{P}} [M_{t+1} \cdot P_{t+1,\tau-1}]. \quad (\text{A.4})$$

That is, if the market is free of arbitrage, the current price of the bond is the expectation of the discounted future price. The pricing kernel is assumed to be a function of the short rate and the risk perceived by the market:

$$M_{t+1} = \exp(-i_t - \frac{1}{2} \Lambda_t' A_t - \Lambda_t' \epsilon_{t+1}^{\mathbb{P}}). \quad (\text{A.5})$$

According to Ang and Piazzesi (2003), under the arbitrage-free assumption, it is guaranteed the existence of an equivalent risk-neutral measure (also called martingale measure) $\mathbb{Q}$, such that the price of the bond for any $t+1$ satisfies:

$$P_{t,\tau} = E_t^{\mathbb{Q}} [\exp(-i_t) \cdot P_{t+1,\tau-1}]. \quad (\text{A.6})$$

The state dynamics under the risk-neutral $\mathbb{Q}$ -measure is given by

$$X_t = \mu^{\mathbb{Q}} + \Phi^{\mathbb{Q}} X_{t-1} + \Sigma \epsilon_t^{\mathbb{Q}}, \epsilon_t^{\mathbb{Q}} \sim N(0, I), \quad (\text{A.7})$$

where the transition equations under both measures are related such that: $\mu^{\mathbb{Q}} = \mu^{\mathbb{P}} - \Sigma \lambda_0$ and $\Phi^{\mathbb{Q}} = \Phi^{\mathbb{P}} - \Sigma \lambda_1$ (see, for example, Nyholm, 2018, p. 7).

Next, in order to derive a closed form solution, we assume that the price follows an exponential affine function of the state variables:

$$P_{t,\tau} = \exp(A_{\tau} + B_{\tau}' X_t). \quad (\text{A.8})$$

Consequently, yields are affine on the states:

$$y_{t,\tau} = -\frac{1}{\tau} \cdot \log(P_{t,\tau}) = -\frac{A_{\tau}}{\tau} - \frac{B_{\tau}'}{\tau} X_t. \quad (\text{A.9})$$

To find A_{τ} and B_{τ} , we first insert the pricing Eq. (A.8) on both sides of Eq. (A.6):

$$\exp(A_{\tau} + B_{\tau}' X_t) = E_t^{\mathbb{Q}} [\exp(-i_t + A_{\tau-1} + B_{\tau-1}' X_{t+1})]. \quad (\text{A.10})$$

Then, we include Eqs. (A.2) and (A.7) in the equation above:

$$\begin{aligned} \exp(A_{\tau} + B_{\tau}' X_t) &= \exp(-\rho_0 - \rho_1' X_t + A_{\tau-1} + B_{\tau-1}' \mu^{\mathbb{Q}} + B_{\tau-1}' \Phi^{\mathbb{Q}} X_t) \cdot E_t^{\mathbb{Q}} \left[\exp \left(B_{\tau-1}' \Sigma \epsilon_{t+1}^{\mathbb{Q}} \right) \right] \\ &= \exp \left(-\rho_0 - \rho_1' X_t + A_{\tau-1} + B_{\tau-1}' \mu^{\mathbb{Q}} + B_{\tau-1}' \Phi^{\mathbb{Q}} X_t + \frac{1}{2} B_{\tau-1}' \Sigma \Sigma' B_{\tau-1} \right). \end{aligned} \quad (\text{A.11})$$

Finally, taking the log from both sides yields

$$A_{\tau} + B_{\tau}' X_t = -\rho_0 - \rho_1' X_t + A_{\tau-1} + B_{\tau-1}' \mu^{\mathbb{Q}} + B_{\tau-1}' \Phi^{\mathbb{Q}} X_t + \frac{1}{2} B_{\tau-1}' \Sigma \Sigma' B_{\tau-1}. \quad (\text{A.12})$$

Matching coefficients results in the two difference equations:

$$A_{\tau} = -\rho_0 + A_{\tau-1} + B_{\tau-1}' \mu^{\mathbb{Q}} + \frac{1}{2} B_{\tau-1}' \Sigma \Sigma' B_{\tau-1}, \quad (\text{A.13})$$

$$B_{\tau} = (\Phi^{\mathbb{Q}})' B_{\tau-1} - \rho_1 = - \left[\sum_{k=0}^{\tau-1} (\Phi^{\mathbb{Q}})^k \right]' \cdot \rho_1. \quad (\text{A.14})$$

Under no-arbitrage, we derive the initial condition, from the fact that in the end period, the price of the bond is equal to the face value: $P_0 = 1$. It follows the initial conditions: $A_0 = B_0 = 0$.

The key difference between the standard DNS models and its arbitrage-free versions is the fact that there is no constant term; thus: $A_{\tau} = 0$. Moreover, we do not make any assumption regarding the “real” dynamics of the model, since this is not necessary. In fact, as pointed out by Diebold and Rudebusch (2013), the choice of the risk neutral parameters does not constrain the choice of the parameters and dynamics under the probability measure $\mathbb{P}$. This gives us the freedom to specify the “real” dynamics that the state factors follow in the most appropriate way.

Carrying on with the derivation, we need to choose Φ^Q that leads to the desired loadings. Nyholm (2018) chooses the loadings which approximately match (Söderlind & Svensson, 1997). He uses following matrix structure:

$$\Phi^Q = \begin{pmatrix} 1 & 1-\gamma & 1-\gamma & 1-\gamma \\ 0 & \gamma & \gamma-1 & \gamma-1 \\ 0 & 0 & \gamma & \gamma-1 \\ 0 & 0 & 0 & \gamma \end{pmatrix}. \quad (\text{A.15})$$

Inserting it in Eq. (A.14) results in

$$B_\tau = \begin{pmatrix} \tau & & & \\ \tau - \frac{1-\gamma^\tau}{1-\gamma} & & & \\ -\tau\gamma^{\tau-1} + \frac{1-\gamma^\tau}{1-\gamma} & & & \\ -\frac{1}{2}\tau(\tau-1)(\gamma-1)\gamma^{\tau-2} & & & \end{pmatrix}. \quad (\text{A.16})$$

Finally, following Joslin et al. (2011), further restrictions are imposed to achieve exact identification: $\mu^Q = (\mu_i^Q \ 0 \ 0 \ 0)$ and $\rho_0 = 0$, in which case the adjustment term from Eq. (A.13) reduces to

$$A_\tau = A_{\tau-1} + \tau\mu_i^Q + \frac{1}{2}B_{\tau-1}'\Sigma\Sigma'B_{\tau-1}. \quad (\text{A.17})$$

This specification guarantees the maximal parsimony of the model, with the only difference to the DNS being the additional parameter μ_i^Q .

Once the no-arbitrage model is derived, we can now turn to the question of how to include unspanned variables in the model following Li et al. (2011). Let X_t^f be a vector containing the four latent factors and X_t^s be the vector of observable variables (in our case, the three sentiment scores). As before, the short rate follows an affine form of state variables:

$$i_t = \rho_0 + (\rho_1^f)'X_t^f + (\rho_1^s)'X_t^s. \quad (\text{A.18})$$

Furthermore, we set

$$\rho_1^f = (1 \ 0 \ 0 \ 0)', \quad (\text{A.19})$$

$$\rho_1^s = 0. \quad (\text{A.20})$$

Under the measure $\mathbb{Q}$, the state variables follow a first-order VAR process:

$$\begin{pmatrix} X_t^f \\ X_t^s \end{pmatrix} = \mu^Q + \begin{pmatrix} \Phi_f^Q & 0 \\ \Phi_{fs}^Q & \Phi_s^Q \end{pmatrix} \begin{pmatrix} X_{t-1}^f \\ X_{t-1}^s \end{pmatrix} + \epsilon_t^Q, \quad (\text{A.21})$$

where the transition matrix for the unobservable variables is equal to before:

$$\Phi_f^Q = \begin{pmatrix} 1 & 1-\gamma & 1-\gamma & 1-\gamma \\ 0 & \gamma & \gamma-1 & \gamma-1 \\ 0 & 0 & \gamma & \gamma-1 \\ 0 & 0 & 0 & \gamma \end{pmatrix}. \quad (\text{A.22})$$

Hence, the price of the zero-coupon bond also remains the same:

$$P_{t,\tau} = \exp(A_\tau + B_\tau^{f'}X_t^f). \quad (\text{A.23})$$

From Eq. (A.14), we know that:

$$\begin{pmatrix} B_\tau^f \\ B_\tau^s \end{pmatrix} = \begin{pmatrix} \Phi_f^{Q'} & \Phi_{fs}^{Q'} \\ 0 & \Phi_s^{Q'} \end{pmatrix} \begin{pmatrix} B_{\tau-1}^f \\ B_{\tau-1}^s \end{pmatrix} - \begin{pmatrix} \rho_1^f \\ 0 \end{pmatrix}. \quad (\text{A.24})$$

Therefore, it follows

$$B_\tau^s = (\Phi_s^Q)' B_{\tau-1}^s = 0, \quad \text{for } \tau > 0 \quad (\text{A.25})$$

$$B_\tau^f = (\Phi_f^Q)' B_{\tau-1}^f - \rho_1^f = - \left[\sum_{k=0}^{\tau-1} (\Phi_f^Q)^k \right]' \cdot \rho_1^f. \quad (\text{A.26})$$

To simplify the notation, the variance-covariance matrix is defined as $\Sigma\Sigma' = \Omega$. Therefore, we find the adjustment term by letting Ω_f be the upper left block of the variance-covariance matrix corresponding to the factor innovations in the state dynamics. Since $B_\tau^s = 0$, we get

$$B_\tau' \Omega B_\tau = B_\tau^{f'} \Omega_f B_\tau^f. \quad (\text{A.27})$$

Thus, the adjustment term is

$$A_\tau = -\rho_0 + A_{\tau-1} + B_{\tau-1}^{f'} \mu^Q + \frac{1}{2} B_{\tau-1}^{f'} \Omega_f B_{\tau-1}^f. \quad (\text{A.28})$$

It is important to notice that Φ_s^Q and Φ_{fs}^Q are not identified, in which case the price of risk parameters λ_0 and λ_1 are also not identified. However, this is not a concern since the only purpose of identification of the mean reversion matrix under the measure $\mathbb{Q}$ is for the latent factors to enter the yield curve according to Söderlind and Svensson (1997).

The final step is to include the monetary policy rule in the transition matrix under the $\mathbb{P}$ -measure. Based on Nyholm (2018), we assume the following structure, where the first row specifies the monetary policy rule, rows 2 to 4 allow for interaction between the yield curve factors, and rows 5 to 9 represent the autoregressive processes for sentiment:

$$\Phi^{\mathbb{P}} = \begin{pmatrix} \phi_{(1,1)} & 0 & 0 & 0 & \phi_{(1,5)} & \phi_{(1,6)} & \phi_{(1,7)} \\ \phi_{(2,1)} & \phi_{(2,2)} & \phi_{(2,3)} & \phi_{(2,4)} & 0 & 0 & 0 \\ \phi_{(3,1)} & \phi_{(3,2)} & \phi_{(3,3)} & \phi_{(3,4)} & 0 & 0 & 0 \\ \phi_{(4,1)} & \phi_{(4,2)} & \phi_{(4,3)} & \phi_{(4,4)} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \phi_{(5,5)} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \phi_{(6,6)} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \phi_{(7,7)} \end{pmatrix} \quad (\text{A.29})$$

A.2. Estimation methods

We first estimate the Sentiment Taylor rule under 1 and 2 regime using Ordinary Least Square (OLS). To estimate the optimal threshold of the TAR model, $\hat{c}$, we execute a grid-search, whereby the highest and lowest 15% of the values in the sample are excluded to avoid an extremely small sample size, as suggested by Enders (2015, p. 428).

The estimation of the DNS is achieved by the two-step method motivated by Diebold and Li (2006):

Step 1: The static Nelson–Siegel model is fitted for each period $t = 1 \dots T$ using OLS. This delivers the three-dimensional time series of estimated factors and a corresponding N -dimensional series of residual pricing errors

$$\hat{X}_t = \hat{B}^{-1} y_t', \quad (\text{A.30})$$

$$\hat{\epsilon}_t = y_t - \hat{B} \cdot \hat{X}_t, \quad (\text{A.31})$$

where $\hat{B}$ is the estimated matrix containing Nelson–Siegel loadings and $\hat{B}^{-1}$ is its pseudo-inverse. Moreover, the parameter γ is chosen via grid-search, such that the sum of squared residuals is minimized

$$\hat{\gamma} = \arg \min_{\gamma} \sum_{t=1}^T \sum_{\tau=1}^N \epsilon_t(\tau)^2. \quad (\text{A.32})$$

Step 2: The coefficients of the first-order VAR process are estimated via OLS, which, under the assumptions of serially uncorrelated error terms with constant variance is consistent and asymptotically efficient (see, for example, Enders, 2015, p. 290).

As pointed out by Diebold and Rudebusch (2013, p. 57) “the cost of two-step estimation is its possible statistical suboptimality, insofar as the first-step parameter estimation error is ignored in the second-step, which may distort second-step inference”. However, in this paper, we choose this method for the sake of simplicity and numerical stability.

Finally, as noted above, a big challenge in the application of no-arbitrage models concerns estimation and inference. According to Li et al. (2011), even though Maximum Likelihood Estimation (MLE) methods are suitable for state-space models with latent variables, finding a global optimum with a high dimension of parameters, typical in multi-factor term structure models, can be challenging. Based on the work of Joslin et al. (2011), which shows that the maximum likelihood function can be partitioned such that the $\mathbb{Q}$ - and $\mathbb{P}$ -measure

Fed's Mester: Fed Will Need To Raise Rates Three Times In 2018, 2019 January 18, 2018 18:06 ET (23:06 GMT)
Federal Reserve Bank of Cleveland President Loretta Mester said Thursday she anticipated the Fed would need to raise its benchmark interest rate three times in 2018 and 2019. In a speech in New York, Ms. Mester said the economy had finally returned to normal after the last recession, leaving Fed officials with the responsibility of keeping it from overheating. "If the economy evolves as I anticipate, we'll need to make some further increases in interest rates this year and next year, at a pace similar to last year's," she said. In 2017, the Fed raised rates three times to a range of between 1.25% and 1.5%. Officials have penciled in three more rate increases this year and two next year. "The economy is coming into 2018 with positive momentum, and for the first time in a while, there are some more salient upside risks to the forecast," Ms. Mester said. Raising rates gradually "gives inflation time to move back to goal while, at the same time, avoiding a buildup of risks to macroeconomic stability that could arise if the economy is allowed to overheat, with the Fed then having to raise rates sharply in response," she said. Inflation has fallen short of the Fed's 2% target for most of the past five years, but Ms. Mester said she expects to hit the goal in the next year or two. Ms. Mester said she expected fiscal policy to add another 0.25% to 0.5% to economic growth this year and next but "the effect could be larger." Much of that additional growth will come from household spending, which is expected to get a boost from lower personal tax rates, she said. But it remains to be seen whether the tax overhaul causes businesses to pick up investment and hiring, she added. "The majority of our business contacts have told us that while they welcome lower tax rates, they aren't planning to make significant changes to their capital or hiring plans as a result of the change in taxes," she said. Because of the uncertainty surrounding the new law's effects, Ms. Mester said she didn't factor possible productivity increases or higher labor-force participation rates from the tax law into her long-term forecasts.

Fig. B.4. Construction of interest rate sentiment scores: An example.

News Article used to generated the interest rate sentiment scores in Table 2. The article was extracted by matching Ravenpack data with the Dow Jones Newswire database using timestamps.

parameters can be estimated separately, Nyholm (2018) developed a stepwise estimation method to estimate the AFSR as follows:

Step 1: Taking $\hat{\gamma}$ as given, a closed-form solution for $\hat{\phi}_f^Q$ can be found using Eq. (A.22).

Step 2: The yield curve factors X_t^f can then be estimated via OLS:

$$\hat{X}_t^f = (\hat{B}^f)^{-1} y_t^f, \quad (\text{A.33})$$

where $B^f = -B_\tau^f / \tau$ (see).¹³

Step 3: As before, in order to estimate $\hat{\gamma}$, a grid-search is executed, such that the sum of squared errors is minimized (see Eq. (A.32)).

Step 4: The dynamics under the $\mathbb{P}$ -measure can be found using the extracted factors. Notice that due to the parameter constraints, OLS is not the most efficient method. In this case, Enders (2015, p. 303) recommends the use of Seemingly Unrelated Regression (SUR), which yields an efficient residual covariance matrix (see, for example, Wooldridge, 2010, p. 185).

Step 5: Lastly, we use the estimated factors and the residual covariance matrix to estimate μ_t^Q :

$$\hat{\mu}_t^Q = \arg \min_{\mu_t^Q} \sum_{t=1}^T \sum_{\tau=1}^N \left[y_t - \hat{A} - \hat{B}^f \cdot \hat{X}_t^f \right]^2$$

where $A = -A_\tau / \tau$ (see Eq. (A.28)).

An advantage of the stepwise estimation method is that it makes it convenient and easy to include a regime-switching model. Since only the dynamics under the $\mathbb{P}$ -measure are assumed to change regimes, we can add the estimation process of the TAR to steps 2 and 4 of the DNS and AFSR estimation process, respectively.

Appendix B. Figure appendix

B.1. Further details about sentiment construction

See Figs. B.4–B.9.

B.2. Further empirical results

See Figs. B.10 and B.11.

Appendix C. Table appendix

C.1. Further model details and summary statistics

See Tables C.11–C.13.

C.2. Further forecasting results

See Tables C.14 and C.15.

¹³ See also de Los Rios (2015) for similar estimation steps with principal component weights used to obtain the factors. Yet, given that both models are linear combinations of yields, they are identical up to a rotation (see, for example, Joslin et al., 2011).

Federal Reserve Will Make Interest Rate Cut - Economists October 18, 2002 14:41 ET (18:41 GMT)
The purchasing manager's index and employment numbers, which come out before the Federal Reserve meets Nov. 1, will be enough to convince the Fed to make a quarter percent interest rate cut, said Stuart Hoffman, chief economist PNC Financial Services Group (PNC), in an interview with CNBC Friday. In the same interview, Richard Rippe, chief economist for Prudential Financial Inc. (PRU), agreed that the Federal Reserve would make a cut after the reports. Hoffman said if unemployment remains up and the purchasing manager's index goes down, it will be enough ammunition for the Federal Reserve to make the cut. Hoffman added the economy is a 'mixed picture.' The mixed indicators for the week show a 'two steps forward, one step back' movement for the market. He said the consumer price index numbers that came out earlier Friday were 'benign.' He added the higher trade deficit would drag on the U.S. gross domestic product. Rippe also agreed the housing markets have been the strongest sector by far, although he sees the growth slowing in the fourth quarter with the rising mortgage interest rates. The housing boom will continue to drive consumer spending, he said. 'I think the economy is making progress and not facing a double-dip recession,' Rippe said. Despite general concerns of deflation, the consumer price index numbers indicate minor inflation, he said.

Fig. B.5. Construction of interest rate sentiment scores: An example.
News Article used to generated the interest rate sentiment scores in Table 2. The article was extracted by matching Ravenpack data with the Dow Jones Newswire database using timestamps.

Fed's Quarles Says Low U.S. Inflation Readings Are Not A 'great Concern' February 22, 2018 00:16 ET (05:16 GMT)
Recent low inflation readings are not a "great concern," and the U.S. central bank should continue to raise interest rates gradually, said Randall Quarles, the new Federal Reserve vice chairman for supervision, on Thursday. In a speech in Tokyo, Quarles said the economy "is performing very well" with a strong labor market and "likely only temporary softness in inflation." Quarles noted that headline personal consumption expenditure index rose at a 1.7% annual rate in December, just below the Fed's 2% target. "Suffice to say, a deviation from our target by a few tenths of 1 percentage point, especially one I expect to fade, does not cause me great concern," Quarles said. A Trump appointee, Quarles was optimistic about the impact of the Republican tax cuts. He said the investment drought that has afflicted the U.S. economy over the past five years "may finally be breaking."

Fig. B.6. Construction of inflation sentiment scores: An example.
News Article used to generated the inflation sentiment scores in Table 2. The article was extracted by matching Ravenpack data with the Dow Jones Newswire database using timestamps.

MARKET TALK: US PPI Jumps 1.8% In Nov December 15, 2009 08:38 ET (13:38 GMT)
Surging energy costs drove US wholesale prices up nearly twice as much as expected in Nov while core prices posted their largest increase in more than a year. Nov PPI leapt 1.8%, after rising by an unrevised 0.3% in Oct. Economists had expected prices would climb by 1.0%. The data show the PPI is up 2.4% from Nov 2008. Core PPI, which excludes food and energy, rose 0.5% in Nov, driven up by higher prices for light motor trucks and cigarettes. The increase in core PPI is the largest since Oct 2008 and is more than double the 0.2% rise economists had expected."

Fig. B.7. Construction of inflation sentiment scores: An example.
News Article used to generated the inflation sentiment scores in Table 2. The article was extracted by matching Ravenpack data with the Dow Jones Newswire database using timestamps.

Fed's Kaplan: Texas Unemployment Lowest Since '70s' January 10, 2018 12:54 ET (17:54 GMT)
Texas labor market is firing on all cylinders, Dallas Fed President Robert Kaplan says, pointing to 2.5% job growth last year and another likely 3% this year. "The unemployment rate right now in the state of Texas is approximately 3.8%, and that is the lowest recorded rate of unemployment – believe it or not – since the 70s," Kaplan tells an audience at SMU in Dallas. While the Texas economy collapsed in the 1980s oil bust, it's handling the 2014 bust better, he says, noting a spike in Permian Basin oil drilling. "Texas is growing in population, we're a magnet, have been and will continue to be based on everything I see, for people and firms.

Fig. B.8. Construction of unemployment sentiment scores: An example.
News Article used to generated the unemployment sentiment scores in Table 2. The article was extracted by matching Ravenpack data with the Dow Jones Newswire database using timestamps.

DATA SNAP: US Jobless Rate Rises For 1st Time Since Dec April 05, 2002 08:30 ET (13:30 GMT)	
U.S. employers added 58,000 workers to their payrolls in March, ending seven months of cutbacks, the government said in a report that suggested the economy is recovering from its first recession in a decade. Non-farm business payrolls grew for the first time since July, the Labor Department said. The department revised its estimate of payrolls in February, saying they shrank rather than grew. The unemployment rate, meanwhile, rose in March for the first time in three months, climbing two-tenths of a percentage point to 5.7%. Those numbers surprised Wall Street, which expected payrolls to grow 50,000 and the unemployment rate to rise no higher than 5.6%. The report suggested the U.S. economic recovery isn't strong enough to prompt the Federal Reserve to raise interest rates before the summer. Investors are likely to revise their expectations as a result. U.S. economic data released in the last month have convinced most experts that the country's first recession in a decade is over. The economy grew a surprising 1.7% in the fourth quarter, up from an initial estimate of 0.2%. That reflected the fastest increase in consumer spending in nearly two years and the biggest increase in overall corporate profits in more than 50 years. Investors, as a result, have expected the Fed to raise interest rates rapidly - beginning no later than June. Futures contracts on the Fed's key federal funds rate imply an expectation that the rate will climb to 3.5% by the end of the year, up from the current level of 1.75%. Fed policymakers, however, have sought to lower those expectations, saying the recovery remains fragile. 'The preconditions for a recovery are in place,' William McDonough, president of the Federal Reserve Bank of New York, told the National Association for Business Economics last week. 'But not all indicators are flashing green.' Job creation, he said, has been sluggish so far, which implies it will 'take some time' for the country to recover the 1.5 million jobs lost during the recession last year. Economists say the Fed never raises interest rates until after the unemployment rate has stabilized in the wake of a recession. Many forecasters say the unemployment rate may rise to 6% this year. The Labor Department attributed most of the increase in payrolls in March to job growth in the services industry. The services industry added 118,000 jobs, the biggest increase in 18 months. Lois Orr, a Labor Department official, said the numbers indicated the industry has 'nearly recovered' from 245,000 job losses late last year. The services-producing industry added 135,000 jobs. The manufacturing industry, however, trimmed 38,000 jobs in March, the smallest cutback since late 2000. In the year that ended in January, the manufacturing industry lost an average of 111,000 jobs a month, the Labor Department said. The employment report suggested that inflation risks remained tame despite the flurry of tax and interest-rate cuts over the last year. Average hourly earnings rose four cents, or 0.3%, to \$14.67 in March after rising by the same amount in February. Over the last 12 months, average hourly wages have risen 3.5%. But the average work week held steady: In March, the week lasted 34.2 hours. The government revised its estimates of growth in payrolls in February. Payrolls declined by 2,000 in February, down from the initial estimate of a 66,000 increase. A Labor Department spokeswoman said a revision of that size isn't unusual.	

Fig. B.9. Construction of unemployment sentiment scores: An example.

News Article used to generated the unemployment sentiment scores in Table 2. The article was extracted by matching Ravenpack data with the Dow Jones Newswire database using timestamps.

Table C.11

Results of short rate regression on sentiment using first differences.

	(1)	(2)	(3)
α	0.041 (0.064)	0.017 (0.060)	0.023 (0.060)
δ_1	-0.037*** (0.045)	-0.023 (0.043)	-0.027 (0.042)
δ_2	-0.002 (0.001)	0.0002 (0.001)	-0.0001 (0.001)
θ_1		-0.083*** (0.014)	-0.084*** (0.013)
θ_2		0.006 (0.011)	
θ_3		0.010 (0.016)	
Observations	244	244	244
Adjusted R ²	0.022	0.189	0.193
AIC	-124.3	-167.3	-170.2
BIC	-110.3	-142.8	-152.7

The table shows the regression of the first difference of the U.S. 3-month Treasury yield on different variables. Column (1) uses inflation and unemployment rate, column (2) adds the three sentiment scores, and column (3) omits inflation and unemployment rate sentiment. Heteroskedastic-consistent standard errors are shown in parentheses. Sentiment scores are standardized. ***, ** and * denote significance at the 1%, 5%, and 10% level, respectively.

Table C.12

Results for sentiment-based regime-switching model using first differences.

	α	δ_1	δ_2	θ_1	θ_2	θ_3	c
Regime 1 $S_{i,t} > c$	-0.019 (0.388)	-0.189 (0.193)	0.007 (0.006)	0.069 (0.086)	-0.041 (0.072)	0.054 (0.081)	0.656 (0.442)
Regime 2 $S_{i,t} \leq c$	-0.038 (0.040)	0.011 (0.027)	-0.000 (0.001)	-0.067*** (0.012)	0.003 (0.008)	0.015 (0.017)	
Observations	Adjusted R ²	AIC	BIC				
244	0.261	-186.2	-140.8				

The table shows the results of the regime-switching model based on the threshold auto-regressive model. The threshold is based on interest rate sentiment. The first row shows the coefficients of the regime 1, which represents the high-sentiment regime. The second row presents the coefficients of regime 2, which stands for the low-sentiment regime. Heteroskedastic-consistent standard errors are shown in parentheses. The standard error for the threshold coefficient is estimated using blockwise bootstrapping. Sentiment scores are standardized. ***, ** and * denote significance at the 1%, 5%, and 10% level, respectively.

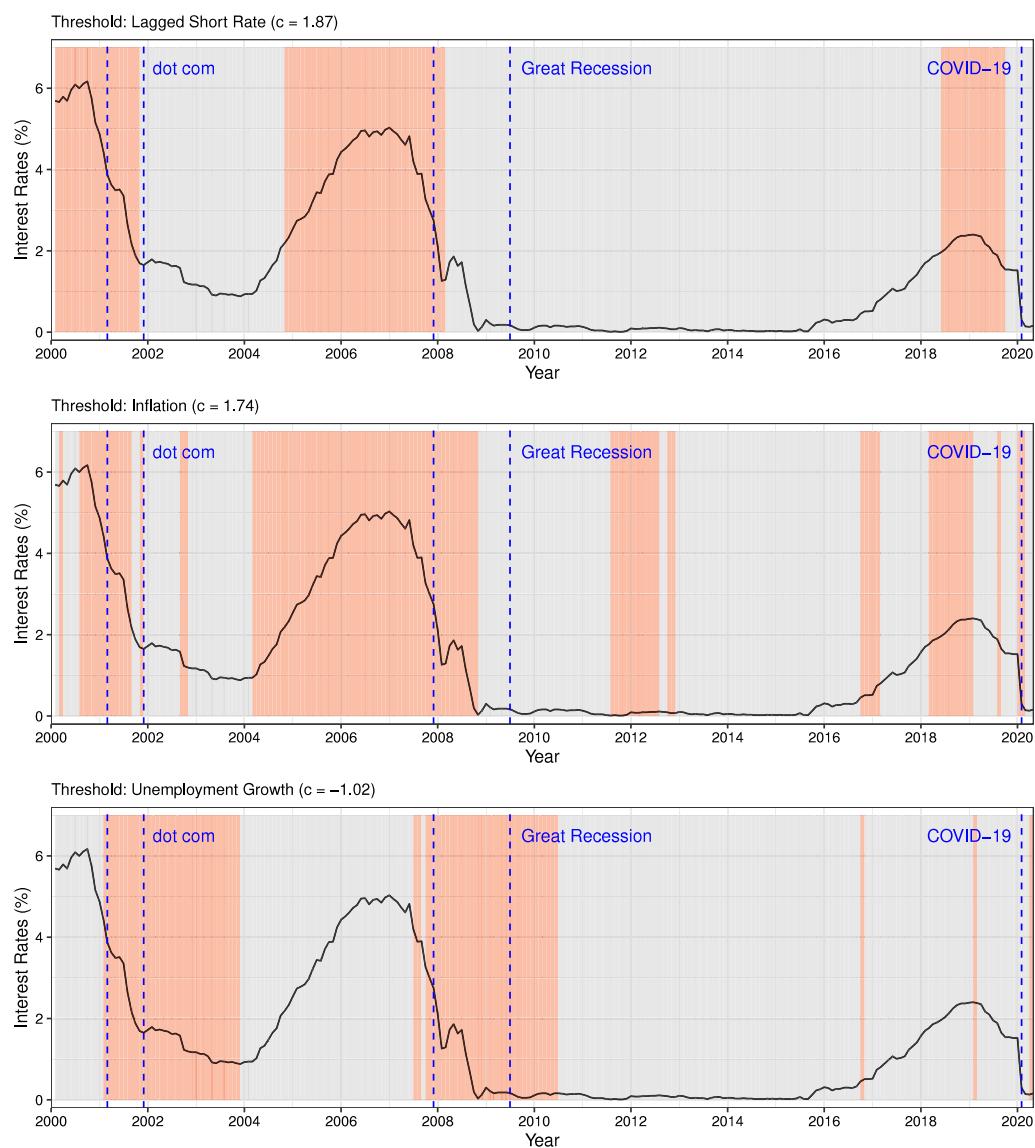


Fig. B.10. U.S. 3-month treasury yield: regime changes with alternative variables.

The figure shows monthly development of the U.S. 3-month Treasury yield from January 2000 to May 2020. The red shaded area represents the period characterized by high values of the variable, while the gray shaded area represents low values of the variable. The threshold parameters are shown in parentheses. The dashed blue lines indicate the periods of economic contractions according to NBER, which are indicated in blue.

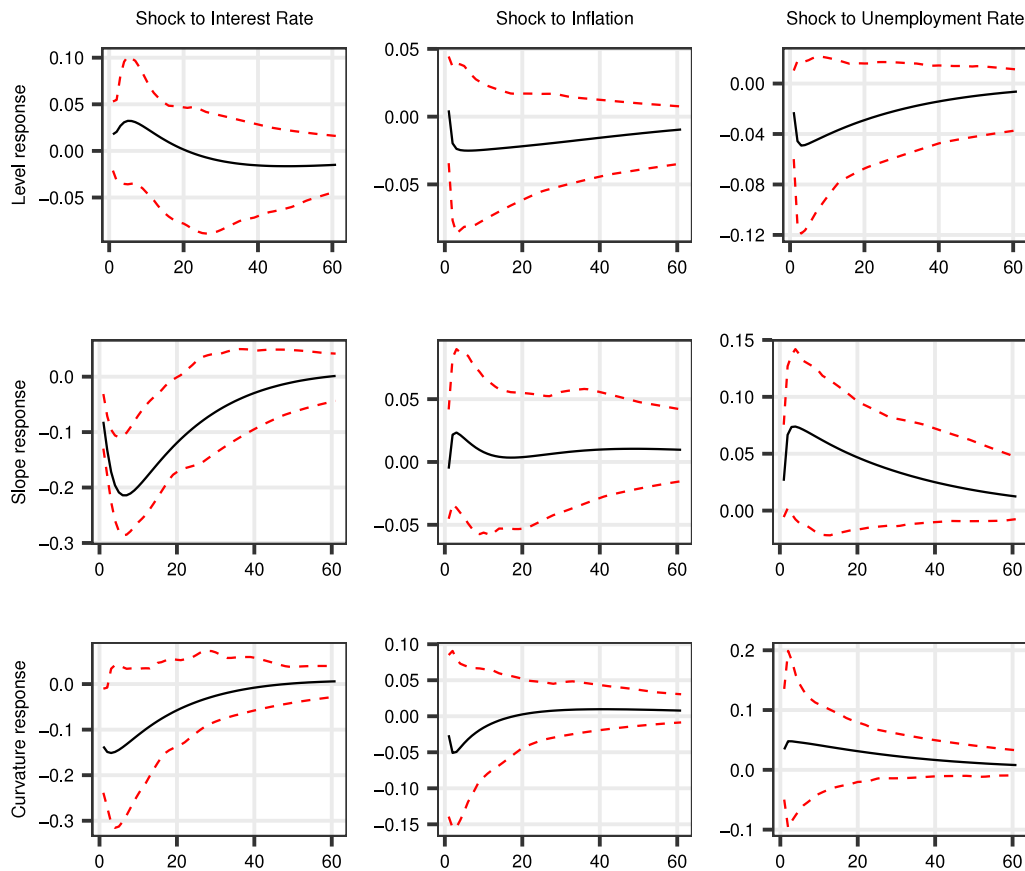


Fig. B.11. Impulse response functions of yield curve factors to sentiment shocks.

The figure displays the impulse response functions of yield curve factors to sentiment shocks using the Dynamic Nelson–Siegel Model. The black line represents the impulse response path for 0 to 60 periods ahead, while the red dotted lines represent the 90% confidence intervals.

Table C.13

Parameter estimates for regime-switching yield curve models using first differences.

Panel A: Level Factor from Dynamic Nelson–Siegel Model							
	μ	S_{t-1}	C_{t-1}	$S_{t,d-1}$	$S_{\pi,t-1}$	$S_{u,t-1}$	c
Regime 1	0.316	−0.034	0.109**	−0.057	−0.147	0.021	0.403
$S_{i,t} > c$	(0.179)	(0.027)	(0.049)	(0.074)	(0.102)	(0.059)	(0.236)
Regime 2	0.172	0.017	−0.010	0.040	0.014	−0.053	
$S_{i,t} \leq c$	(0.106)	(0.016)	(0.019)	(0.036)	(0.033)	(0.049)	
Observations	Adjusted R ²	AIC	BIC				
244	0.071	192.8	238.3				
Panel B: Short Rate Factor from Arbitrage-free Short Rate based Model							
	μ	S_{t-1}	C_{t-1}	$S_{t,d-1}$	$S_{\pi,t-1}$	$S_{u,t-1}$	c
Regime 1	−0.560	–	–	0.160	−0.063	0.052	0.887
$S_{i,t} > c$	(0.206)			(0.092)	(0.094)	(0.061)	(0.559)
Regime 2	0.008	–	–	−0.061***	0.004	0.005	
$S_{i,t} \leq c$	(0.012)			(0.014)	(0.008)	(0.015)	
Observations	Adjusted R ²	AIC	BIC				
244	0.887	−100.6	−69.18				

Panel A presents the regression results for the threshold autoregressive Dynamic Nelson–Siegel model. Panel B shows the results for the threshold autoregressive Arbitrage-Free Short Rate based model. The former includes a sentiment-based threshold in the level process, while the latter includes it in the short rate process. Standard errors are shown in parentheses. The standard error for the threshold coefficient is estimated using blockwise bootstrapping. Sentiment scores are standardized. ***, ** and * denote significance at the 1%, 5%, and 10% level, respectively.

Table C.14

Out-of-sample forecast using sentiment in the dynamic Nelson–Siegel model starting in 2016.

	1-Month		2-Months		3-Months		6-Months		12-Months	
	1 Regime	2 Regimes	1 Regime	2 Regimes	1 Regime	2 Regimes	1 Regime	2 Regimes	1 Regime	2 Regimes
3-Months Yield										
Taylor Rule										
(1)	20.41**	45.33*	29.40**	44.12**	37.35**	51.77**	46.31*	66.43	71.54**	99.69
(2)	19.43**	41.80	29.00**	44.74*	36.04**	53.45	41.44**	68.41	69.81**	100.73
(3)	19.27**	43.35*	28.78**	42.97**	35.71**	50.47*	40.87**	63.90	68.48**	98.16
AR(1)										
(1)	20.13**	31.50*	32.37**	41.27*	42.45**	50.20	59.52	67.02	84.87**	98.32
(2)	18.74**	27.40*	29.32**	38.16*	37.80**	48.38	49.12	64.57	79.67*	98.53
(3)	18.51**	25.92**	28.93**	35.82**	37.27**	44.77*	48.62	60.05	78.18**	95.88
Random Walk	20.07**		32.17**		42.08**		59.31		96.87	
2-Years Yield										
Taylor Rule										
(1)	17.98*	45.23*	28.53**	45.09*	38.68**	52.96*	58.26**	68.60**	92.11**	102.64**
(2)	17.03**	42.94	27.23**	45.76*	36.55**	53.86	54.86**	70.21	91.22**	101.98**
(3)	16.85**	43.71*	26.93**	44.19*	36.16**	51.70*	54.00**	66.57**	90.15**	100.87**
AR(1)										
(1)	18.43*	34.10*	29.95**	40.99	40.87*	49.87	63.07**	67.82*	97.02**	101.32**
(2)	17.03**	31.30*	27.47**	38.98	37.29**	48.44	57.07**	66.55**	94.09**	100.57**
(3)	16.90**	29.75*	27.17**	36.83*	36.88**	45.46	56.24**	62.74**	92.86**	99.59**
Random Walk	17.92*		28.56**		38.82**		59.87**		100.24**	
5-Years Yield										
Taylor Rule										
(1)	21.59**	44.65*	34.30**	47.79*	45.64**	55.98**	69.62**	71.25**	103.56**	99.65**
(2)	21.35**	43.51*	33.16**	48.75*	44.08**	56.97*	67.62**	72.99**	103.25**	98.79**
(3)	20.99**	43.58*	32.71**	47.14**	43.55**	55.06**	66.63**	69.95**	102.48**	98.23**
AR(1)										
(1)	21.17**	35.77*	33.53**	43.38*	44.43**	52.22*	67.30**	69.25**	101.19**	97.69**
(2)	20.74**	34.15*	32.53**	42.46*	43.01**	51.61*	65.18**	68.88**	100.44**	96.55**
(3)	20.54**	33.07*	32.31**	41.05*	42.74**	49.67**	64.32**	66.17**	99.54**	96.93**
Random Walk	20.12**		31.13**		40.76**		61.09**		95.63**	
8-Years Yield										
Taylor Rule										
(1)	23.21*	45.07*	36.60**	49.02**	48.86**	56.94**	74.84**	70.29**	111.48*	94.51**
(2)	23.56*	44.91*	36.21**	50.61*	48.29**	58.55*	73.83**	72.51**	111.31*	93.87**
(3)	23.12**	44.67*	35.69**	48.95**	47.70**	56.63**	72.91**	69.62**	110.93*	93.53**
AR(1)										
(1)	21.82**	36.65*	34.40**	44.22*	45.54**	52.76**	67.88**	67.41**	103.42**	91.73**
(2)	22.29**	35.91*	34.80**	44.06*	45.92**	52.81**	67.92**	67.51**	103.77**	90.38**
(3)	22.12**	35.10*	34.64**	43.00*	45.75**	51.36**	67.25**	65.35**	103.28**	91.50**
Random Walk	20.47**		31.24**		40.52**		57.96**		87.83**	
10-Years Yield										
Taylor Rule										
(1)	23.45*	45.40*	37.03**	49.36**	49.69**	56.90**	76.59**	69.04**	115.10	91.29*
(2)	24.11*	45.61	37.09**	51.22*	49.64**	58.81*	76.05**	71.49**	114.94	90.81**
(3)	23.62*	45.38	36.54**	49.64**	49.04**	56.96**	75.20**	68.70**	114.76	90.57**
AR(1)										
(1)	21.74**	36.85*	34.22**	44.23*	45.39**	52.44**	67.39**	65.66**	104.23*	88.04**
(2)	22.63*	36.53	35.30**	44.40*	46.61**	52.77**	68.39**	65.96**	104.98*	86.62**
(3)	22.49*	35.84	35.17**	43.52*	46.50**	51.54**	67.84**	64.03**	104.72*	88.03**
Random Walk	20.37**		30.79**		39.76**		55.43**		83.04**	

The table presents the root mean square error (RMSE) of 1-, 2-, 3-, 6-, and 12-month ahead forecasts for the U.S. Treasury yield curve from January 2000 to May 2020 estimated with the Dynamic Nelson–Siegel Model. The models are re-estimated each month, with the sample size increasing by one observation point each time, starting in January 2016. The models with the lowest RMSE for a particular maturity are shown in bold. * and ** indicate the models in the 90% and 75% superior confidence set, respectively.

Table C.15

Out-of-sample forecast using sentiment in the arbitrage-free short rate based model starting in 2016.

	1-Month		2-Months		3-Months		6-Months		12-Months	
	1 Regime	2 Regimes	1 Regime	2 Regimes	1 Regime	2 Regimes	1 Regime	2 Regimes	1 Regime	2 Regimes
3-Months Yield										
Taylor Rule										
(1)	21.87**	20.78**	33.03**	32.49**	42.31**	41.93**	52.28**	56.34	79.36**	94.48
(2)	20.59**	21.94**	31.84**	32.70**	38.85**	41.96**	41.07**	56.83*	77.93**	93.43
(3)	20.44**	21.14**	31.60**	31.94**	38.57**	41.12**	40.76**	55.97*	76.66**	92.48
AR(1)										
(1)	21.13**	19.76**	33.61	31.35**	43.64	41.09**	61.00	54.90	97.92	94.02
(2)	20.28**	21.51**	31.28**	32.52**	38.68**	42.09**	43.51**	57.31*	82.91	93.97
(3)	20.03**	19.95**	30.86**	30.31**	38.07**	39.50**	43.10**	53.58	81.34*	91.73
Random Walk	21.17**		33.76		43.96		61.20		97.63	
2-Years Yield										
Taylor Rule										
(1)	21.09**	20.57**	33.23**	31.10**	44.67**	41.22**	60.95**	60.17**	91.50**	100.85*
(2)	18.61**	21.20**	30.48**	30.71**	39.34**	40.42**	52.36**	60.60**	92.28**	100.20*
(3)	18.41**	20.14**	30.17**	30.06**	38.93**	39.83**	51.95**	59.93**	91.02**	99.41*
AR(1)										
(1)	20.87**	19.31**	34.47	30.23**	46.95	40.49**	69.16	59.10	106.47*	100.37*
(2)	18.68**	21.58**	30.74**	31.06**	40.22**	40.93**	54.03**	61.03**	94.73*	100.39*
(3)	18.22**	18.63**	30.08**	28.91**	39.34**	38.56**	53.38**	58.08**	92.99**	98.77*
Random Walk	21.01*		34.71		47.36		69.57		106.49*	
5-Years Yield										
Taylor Rule										
(1)	23.71**	23.37**	36.31*	33.74**	47.68*	43.15**	62.90**	60.03**	88.80**	93.79**
(2)	21.08**	23.84**	33.24**	33.31**	41.81**	42.32**	55.87**	60.68**	89.44**	93.62**
(3)	20.86**	22.81**	32.92**	32.68**	41.37**	41.71**	55.36**	60.00**	88.21**	92.80**
AR(1)										
(1)	23.41**	22.07**	37.19	32.73**	49.47	42.18**	69.69	58.85**	99.42**	92.86**
(2)	21.27**	24.33**	33.73**	33.78**	43.01**	42.93**	57.75**	61.19**	91.07**	93.76**
(3)	20.75**	21.33**	32.99**	31.55**	41.98**	40.40**	56.78**	58.27**	88.91**	91.87**
Random Walk	23.67*		37.60		50.09		70.32		100.09**	
8-Years Yield										
Taylor Rule										
(1)	24.06**	23.61**	36.73*	33.92**	48.09	43.07**	61.94**	57.94**	85.20**	87.39**
(2)	21.27**	23.93**	33.80**	33.41**	42.35**	42.26**	55.49**	58.64**	85.17**	87.37**
(3)	21.07**	22.93**	33.50**	32.82**	41.92**	41.61**	54.97**	57.93**	84.06**	86.55**
AR(1)										
(1)	23.78**	22.46**	37.61*	32.95**	49.71	42.09**	67.82	56.72**	92.04**	86.19**
(2)	21.48**	24.45**	34.37**	33.90**	43.73**	42.94**	57.82**	59.25**	86.51**	87.56**
(3)	21.02**	21.61**	33.69**	31.74**	42.71**	40.35**	56.72**	56.25**	84.11**	85.42**
Random Walk	24.00*		38.10		50.57		69.38*		95.93**	
10-Years Yield										
Taylor Rule										
(1)	23.85**	23.27**	35.97*	33.20**	47.08	42.00**	60.32**	56.04**	82.65**	83.77**
(2)	21.09**	23.09**	33.46**	32.31**	41.82**	41.00**	54.45**	56.57**	82.51**	83.69**
(3)	20.97**	22.17**	33.22**	31.83**	41.41**	40.39**	53.92**	55.91**	81.45**	82.92**
AR(1)										
(1)	23.64**	22.47**	37.24*	32.52**	49.15	41.31**	66.48	55.00**	88.62**	82.53**
(2)	21.24**	23.56**	34.05**	32.78**	43.35**	41.73**	57.15**	57.24**	83.92**	83.91**
(3)	20.96**	21.26**	33.55**	31.02**	42.47**	39.43**	56.10**	54.42**	81.47**	81.75**
Random Walk	23.50**		37.38		49.71		68.05**		93.81**	

The table presents the root mean square error (RMSE) of 1-, 2-, 3-, 6-, and 12-month ahead forecasts for the U.S. Treasury yield curve from January 2000 to May 2020 estimated with the Arbitrage-Free Short Rate based Model. The models are re-estimated each month, with the sample size increasing by one observation point each time, starting in January 2016. The models with the lowest RMSE for a particular maturity are shown in bold. * and ** indicate the models in the 90% and 75% superior confidence set, respectively.

References

- Algaba, A., Ardia, D., Bluteau, K., Borms, S., & Boudt, K. (2020). Econometrics meets sentiment: An overview of methodology and applications. *Journal of Economic Surveys*, 34, 512–547.
- Ang, A., & Piazzesi, M. (2003). A no-arbitrage vector autoregression of term structure dynamics with macroeconomic and latent variables. *Journal of Monetary Economics*, 50, 745–787.
- Antweiler, W., & Frank, M. Z. (2004). Is all that talk just noise? the information content of internet stock message boards. *The Journal of Finance*, 59, 1259–1294.
- Audrino, F. (2006). Tree-structured multiple regimes in interest rates. *Journal of Business & Economic Statistics*, 24, 338–353.
- Bauer, M. D., & Hamilton, J. D. (2018). Robust bond risk premia. *The Review of Financial Studies*, 31, 399–448.
- Bollen, J., Mao, H., & Zeng, X. (2011). Twitter mood predicts the stock market. *Journal of Computer Science*, 2, 1–8.
- Chinn, M., & Kucko, K. (2015). The predictive power of the yield curve across countries and time. *International Finance*, 18, 129–156.
- Christensen, J. H., Diebold, F. X., & Rudebusch, G. D. (2011). The affine arbitrage-free class of Nelson–Siegel term structure models. *Journal of Econometrics*, 164, 4–20.
- Christensen, J. H., & Rudebusch, G. D. (2015). Estimating shadow-rate term structure models with near-zero yields. *Journal of Financial Econometrics*, 13, 226–259.
- Clarida, R., Gali, J., & Gertler, M. (2000). Monetary policy rules and macroeconomic stability: Evidence and some theory. *Quarterly Journal of Economics*, 115, 147–180.
- Credit Suisse (2021). Credit suisse ravenpack artificial intelligence index. <https://indices.credit-suisse.com/content/CSRPAI5E/brochure/21-024%20R-AI-Index-WEB-rev.pdf>. (Accessed 13 October 2023).
- de Los Rios, A. D. (2015). A new linear estimator for gaussian dynamic term structure models. *Journal of Business & Economic Statistics*, 33, 282–295.
- Diebold, F. X., & Li, C. (2006). Forecasting the term structure of government bond yields. *Journal of Econometrics*, 130, 337–364.
- Diebold, F. X., & Rudebusch, G. D. (2013). *Yield curve modeling and forecasting: The dynamic Nelson–Siegel approach*. Princeton University Press.
- Diebold, F. X., Rudebusch, G. D., & Aruoba, S. B. (2006). The macroeconomy and the yield curve: A dynamic latent factor approach. *Journal of Econometrics*, 131, 309–338.
- Edison, H. J. (1997). The reaction of exchange rates and interest rates to news releases. *International Journal of Finance & Economics*, 2, 87–100.
- Eijffinger, S., Mahieu, R., & Raes, L. (2017). Can the fed talk the hind legs off the stock market? *International Journal of Central Banking*, (47), February 2017.
- Enders, W. (2015). *Applied econometric time series* (4th ed.). John Wiley & Sons.
- Erlwein-Sayer, C. (2017). Forecasting sovereign bond spreads with macroeconomic news sentiment. http://senrisk.eu/wp-content/uploads/2018/10/WhitePaper_ForecastingSovereignBondSpreads.pdf. (Accessed 13 October 2023).
- Erlwein-Sayer, C. (2018). Macroeconomic news sentiment: Enhanced risk assessment for sovereign bonds. *Risks*, 6(141).
- Gentzkow, M., Kelly, B., & Taddy, M. (2019). Text as data. *Journal of Economic Literature*, 57, 535–574.
- Goldberg, L. S., & Leonard, D. (2003). What moves sovereign bond markets? the effects of economic news on US and German yields. *Current Issues in Economics and Finance*, 9.
- Gotthelf, N., & Uhl, M. W. (2019). News sentiment: A new yield curve factor. *Journal of Behavioral Finance*, 20, 31–41.
- Gürkaynak, R. S., Sack, B., & Wright, J. H. (2007). The US treasury yield curve: 1961 to the present. *Journal of Monetary Economics*, 54, 2291–2304.
- Hamilton, J. D. (1988). Rational-expectations econometric analysis of changes in regime: An investigation of the term structure of interest rates. *Journal of Economic Dynamics & Control*, 12, 385–423.
- Hansen, P. R., Lunde, A., & Nason, J. M. (2011). The model confidence set. *Econometrica*, 79, 453–497.
- Heston, S. L., & Sinha, N. R. (2017). News vs. sentiment: Predicting stock returns from news stories. *Financial Analysts Journal*, 73, 67–83.
- Hubert, P., & Labondance, F. (2017). Central Bank sentiment and policy expectations. Working paper.
- Joslin, S., Singleton, K. J., & Zhu, H. (2011). A new perspective on gaussian dynamic term structure models. *The Review of Financial Studies*, 24, 926–970.
- Keynes, J. M. (1936). The general theory of interest, employment and money.
- Li, C., Niu, L., & Zeng, G. (2011). A generalized arbitrage-free Nelson–Siegel term structure model with macroeconomic fundamentals. SSRN working paper.
- Lucca, D. O., & Trebbi, F. (2009). Measuring central bank communication: An automated approach with application to fomc statements. NBER Working paper.
- Mitra, G., & Mitra, L. (2011). *The handbook of news analytics in finance*. John Wiley & Sons.
- Nayak, S. (2010). Investor sentiment and corporate bond yield spreads. *Review of Behavioral Finance*, 2, 59–80.
- Nyholm, K. (2018). A flexible short-rate based four factor arbitrage-free term structure model with an explicit monetary policy rule. <https://wrds-www.wharton.upenn.edu/documents/1164/RavenPack-one-sheet.pdf>. (Accessed 13 October 2023).
- Pooter, M. D. (2007). Examining the Nelson–Siegel class of term structure models. Working paper.
- RavenPack News Analytics [RPNA] (2020). Enhance quantitative research with data derived from traditional and social media. <https://wrds-www.wharton.upenn.edu/documents/1164/RavenPack-one-sheet.pdf>. (Accessed 13 October 2023).
- Renault, T. (2020). Sentiment analysis and machine learning in finance: A comparison of methods and models on one million messages. *Digital Finance*, 2, 1–13.
- Sabes, D., & Sahuc, J. G. (2023). Do yield curve inversions predict recessions in the euro area? *Finance Research Letters*, 52, Article 103416.
- Schmeling, M. (2009). Investor sentiment and stock returns: Some international evidence. *Journal of Empirical Finance*, 16, 394–408.
- Söderlind, P., & Svensson, L. (1997). New techniques to extract market expectations from financial instruments. *Journal of Monetary Economics*, 40, 383–429.
- Spyrou, S. (2013). Investor sentiment and yield spread determinants: Evidence from european markets. *Journal of Economic Studies*.
- Tetlock, P. C. (2007). Giving content to investor sentiment: The role of media in the stock market. *The Journal of Finance*, 62, 1139–1168.
- Tetlock, P. C., Saar-Tsechansky, M., & Macskassy, S. (2008). More than words: Quantifying language to measure firms' fundamentals. *The Journal of Finance*, 63, 1437–1467.
- Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data*. The MIT Press.
- Yang, S., Liu, Z., & Wang, X. (2020). News sentiment, credit spreads, and information asymmetry. *The North American Journal of Economics and Finance*, 52, Article 101179.
- Zhu, X., & Rahman, S. (2015). A regime-switching Nelson–Siegel term structure model of the macroeconomy. *Journal of Macroeconomics*, 44, 1–17.